

**ELABORATING A 'NEW KIND OF FAMILY':
FORDIST SOCIAL POLICIES AND RUBBER WORKERS IN THE GLOBAL SOUTH**

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Chapter 1: Fordism, Social Reproduction and the Global Division of Labor

Fordism has been a slippery concept within historical writing over the last 100 years. Although labor historians and economists agree that it is a central moment in both American and global economic history, there is little agreement on what Fordism actually *is*. In 1990, in a talk humorously titled “What in the F—’s name is Fordism”, British sociologist Simon Clarke noted the imprecision in discussions of Fordism and whatever non-Fordist capitalist epoch is supposed to be its successor: “Much energy has been expended in debating the diffuse characterizations of ‘Not-Fordism’. However, much less attention has been paid to the characterization of Fordism.”¹ Even the popular legacy of Henry Ford himself is deeply ambivalent. He was a benefactor who improved the lives of his workers and a tyrant whose Sociological Department spied on their personal lives; he created the modern middle class but also initiated the culture of consumption that is currently eating the planet; he was a humanitarian who founded schools for the his mostly immigrant employees and he was a vociferous anti-Semite. Only one thing is uniformly agreed upon: Fordism represented a genuine break with existing patterns of production in capitalist society; it was a revolution.

The boundaries of this revolution are blurred and ill-defined. Attempts to define it find themselves faced with irreducible contradictions. Daniel Watson identifies two main approaches to Fordism: a narrow view of Fordism as a system of mass production centered around Henry

¹Simon Clarke, “What in the F—’s Name Is Fordism,” in *Fordism and Flexibility: Divisions and Change*, ed. Nigel Gilbert et al. (Palgrave Macmillan UK, 1992), 13.

Ford's innovative assembly line, and a broader view of Fordism as a "phase of capitalist development taking place several decades after the heyday of the Ford Motor Company" that is characterized by a "careful balance in the American economy between mass production and mass consumption and an increase in the wage rate and social provisions for workers".² In this broader view, the Fordist era was initiated by Henry Ford's productive innovations, but was not fully realized until decades later when his policies were generalized to the rest of the industrialized world.

Each of these definitions is unsatisfactory. In the first place, "[t]here was nothing original in either the detail or the general principles which Our Ford applied to the production of motor vehicles,"³ and so it makes little sense to isolate them from their social component. But if Fordism is to be defined by the unique articulation between these general production principles and his social policy, then why have analysts found it "difficult to discover any examples of 'Fordism' in its pure form, even in the United States"?⁴ Inflation quickly caught up to the Five Dollar Day, and the Sociological Department was dissolved in 1920-21, effectively ending Fordist paternalism and morality bonuses.⁵ In the US, the generalization of social provision to workers was not effected by private capital but by the state, with Franklin D. Roosevelt's New Deal. This flew in the face of both Henry Ford's ideology and his policies, as he was a fierce opponent of the welfare state,⁶ and had placed restrictions on his worker's access to his social benefits.

²Daniel Watson, "Fordism: A Review Essay," *Labor History* 60, no. 2 (2019): 2.

³Simon Clarke, "What in the F—'s Name Is Fordism," in *Fordism and Flexibility: Divisions and Change*, ed. Nigel Gilbert et al. (Palgrave Macmillan UK, 1992), 16.

⁴Simon Clarke, "Overaccumulation, Class Struggle and the Regulation Approach," *Capital & Class* 12, no. 3 (1988): 62.

⁵Stephen Meyer III, *The Five Dollar Day: Labor Management and Social Control in the Ford Motor Company, 1908-1921* (State University of New York Press, 1981), 197.

⁶Sidney Fine, "The Ford Motor Company and the N.R.A.," *The Business History Review* 32, no. 4 (1958): 353–85.

There is a sense in which the primary legacy of Fordism is one of stratification. Simple narratives about the Fordist worker and the family wage do not hold up to scrutiny. In Michigan's auto factories in the first decades of the twentieth century, local workers were increasingly divided into two camps. On the one hand, manufacturing tasks were decomposed into minimal parts, rationalized to the extreme and voided of intellectual stimulation. On the other hand, a great class of managers and engineers was cultivated to design the machines and processes that would combine the decomposed labor of others into finished products - following a managerial revolution that Alfred D. Chandler called the 'visible hand' of the market.⁷ Outside the factories, the living conditions of workers were similarly divided. Skilled and semi-skilled workers, mostly white, were transformed into large-scale commodity consumers and inducted into a class of homeowners, while racialized and menial laborers often lived in crowded slums and still relied significantly on informal labor for survival.

Of course, the labor that contributed to the manufacturing of automobiles was not confined to factory work. The operations of the Ford Motor Company were embedded in a global network of production. Raw materials had to be imported - especially rubber which could only be sourced from the Global South. The stratified labor market in industrial Michigan was mirrored by a global division of labor, in which wild rubber harvesters in the Amazon and the Congo, and to a lesser extent the later rubber plantation workers in southeast Asia, occupied an intensely degraded position. The dynamic of wage labor was not dominant in the rubber regions of the world, and when Northern capital intervened in the rubber industry, it created quite different modes of exploitation. These modes of exploitation were accompanied by different

⁷Alfred Chandler, *The Visible Hand: The Managerial Revolution in American Business* (Belknap Press: An Imprint of Harvard University Press, 1977).

patterns of consumption as well. The rubber workers were made to be dependent on the market for the means of survival, and especially on the importation of canned goods and preserved food from the Global North. But they were not consumers of domestic commodities, sheltered by a commodity house and supported by de-commodified reproductive labor. Their precarity was exploited by capital to effect an extension of the market into the private lives of producers on very different terms than in the Global North.

This chapter takes up early labor stratification in the automotive industry and the methods used to create and discipline it. Keeping the complications listed above in mind, I hope to situate early Fordist production within the birth of what Giovanni Arrighi called the “US systemic cycle of accumulation” - the period of historical capitalism covering much of the twentieth century when the United States dominated global capitalist accumulation. According to Arrighi, this cycle of accumulation was distinguished by “an internalization of transaction costs,”⁸ in which firms vertically integrated in order to retain the surplus value otherwise lost through the purchase of inputs. The scope of this integration was explicitly global, as accumulation in the core has historically been dependent on the exploitation of peripheral regions, and as the working conditions in the periphery responded to demands of accumulation in the core. However, the methods of production and labor control closely related to Fordism, such as assembly lines, the decomposition of tasks, and a family wage, were *not* generalized globally. This raises the question of how vertical integration was effected across national lines, or perhaps more accurately, what were the effects of Fordist production practices in the core on labor conditions in the periphery?

⁸Giovanni Arrighi, *The Long Twentieth Century: Money, Power, and the Origins of Our Times*, 2nd ed. (Verso, 2010), 247.

The second purpose of this chapter is to introduce the dynamics of social reproduction within this global context. Linking these two concepts - the global scope of early Fordism and its regime of social reproduction - is the commodity chain. Wallerstein and Hopkins used the term commodity chain to conceptualize the interconnectedness of global capitalism: “take an ultimate consumable item and trace back the set of inputs that culminated in this item - the prior transformations, the raw materials, the transportation mechanisms, the labor input into each of the material processes, the food inputs into the labor.”⁹ The commodity chain is the axis along which Fordist firms tried to integrate, but this chain includes more than just raw materials and the labor that works them: it is imbricated with “the reproduction of the labor forces involved in these productive activities.”¹⁰ The specific ways in which social reproduction was effected at various points along the Fordist commodity chain will be examined more closely in chapters 2-5, and this chapter is mainly concerned with making the case that the reproductive dynamics in the rubber producing regions of the Global South are not externally articulated with Fordist reproduction, but internal to it. Automotive production required several different types of material and labor inputs, and these correspond to different regimes of social reproduction. In particular, they take advantage of a different articulation between the rationalization of labor, and the expansion of commodity consumption among the workers. The scope of this chapter will focus primarily on the origins of Fordism: the production of automobiles in the first decades of the twentieth century.

⁹Terence K. Hopkins and Immanuel Wallerstein, “Patterns of Development of the Modern World-System,” *Review (Fernand Braudel Center)* 1, no. 2 (1977): 128.

¹⁰Hopkins and Wallerstein, “Patterns of Development,” 128.

The Contradictions of Fordism

In the popular imagination, Fordism comprises two moments of personal genius: Henry Ford's implementation of the automated assembly line and his institution of a family wage with the Five Dollar Day. Faced with the mutually exacerbating problems of rising demand and increasing employee turnover, Ford took the innovative approach of intensifying and de-skilling labor while at the same time increasing its pay.¹¹ These principles were combined in Ford's notorious River Rouge factory, a single complex that took raw materials as inputs turned them into cars. In the meantime, the laborers who worked in this factory were middle class workers who could afford to purchase the commodities they were producing. At one and the same time the efficient production methods transformed automobiles from a luxury good to a near-necessity, and transformed the workers from struggling laborers into participants in the bounty of production. The company's win-win solution to the antagonism of labor and capital transformed American life, and facilitated the dominance of US industry on the world stage. The combination of innovative shop floor management with a reconfiguration of family life ushered in the modern middle class and an era of mass consumerism. In this view, Fordism solved once and for all the antagonism between labor and capital.

This popular conception is problematic on several counts. Ford's production practices were enormously successful, but the core idea of an powered assembly line was borrowed from Chicago slaughterhouses and had first been employed in the automotive industry by a competitor of Ford. In 1911 management at EMF company "installed a mechanized assembly line to boost chassis output to 100 per day."¹² Ford would not install his line (also for chassis production) for

¹¹Carl H. A. Dassbach, "The Origins of Fordism: The Introduction of Mass Production and the Five-Dollar Wage," *Critical Sociology* 18, no. 1 (1991): 77–90.

¹²Wayne Lewchuck, "Fordism and the Moving Assembly Line: The British and American Experience, 1895-1930," in *On The Line: Essays in the History of Auto Work*, ed. Nelson Lichtenstein and Stephen Meyer (University of

another two years, in 1913. Even after Ford's unprecedented success with rationalizing production, his methods would often be taken and improved upon by competitors - most notably General Motors. Ford's real innovation was the extreme to which he took the rationalization of production, and its combination with an aggressive social policy. His famous Five Dollar Day was not a universal wage increase, but a profit share given only to employees whose private lives were approved by agents of the Ford Sociological Association. In addition to investigations into the stability of the workers' families, "the company wanted to know whether or not the workman owned or was purchasing property or a home, whether or not he had a savings account in a bank, and whether or not he was in debt."¹³ In other words, in order to qualify for a family wage, Ford's workers had to demonstrate that they were sufficiently rooted in the community surrounding the factory. When combined with wages that exceeded those offered by competitors, this meant that Ford's prized workers were less able to seek out employment elsewhere. The question was not one of boosting the freedom of workers by increasing their pay, but of limiting them, of compelling them to stay in their jobs.

As we can see, Fordism is a deeply ambiguous concept, full of contradictions. This is part of why it has been studied and written about so extensively - almost any conceivable position on the relationship between labor and capital can find support in the empirical cases of Fordism. But the explanatory power of Fordism as a category in the history of capitalism is exaggerated. Scholarly attempts to resolve the contradictions of Fordism have usually involved a sort of analytical surgery, either removing the social-reproductive aspect and focusing on Henry Ford's specific management practices, or excising Ford almost entirely and treating 'Fordism' as

Illinois Press, 1989), 20.

¹³Stephen Meyer III, *The Five Dollar Day: Labor Management and Social Control in the Ford Motor Company, 1908-1921* (State University of New York Press, 1981), 131.

synecdoche for larger structural changes in the global accumulation of capital in the mid twentieth century. Each of these approaches is untenable. The first reduces Fordism to a loose collection of stolen ideas that were implemented with a ruthless intensity, but which explain very little about the history of capital. The second attempts to establish an ideal type of capitalist society that bears little resemblance to any actually existing world. It inevitably devolves into a list of precursors, exceptions, and modifications that overwhelm the type itself: Taylorism, Sloanism, Toyotism, Keynesianism, Post-Fordism, Neo-Fordism, etc. Sociologist Simon Clarke criticizes this tendency, saying "to explain these variants as failed versions of one ideal-type implies that the latter is in some way more real or more fundamental than the former."¹⁴ Indeed, it is hard to see why Fordism would stand out in this list as the determining element.

The irresolvability of these contradictions are made clear in the work of the regulation school, a group of French scholars who devoted considerable time to defining and delimiting the concept of Fordism. Their scholarship was rigorous, but flawed. Within this group, Michel Aglietta provides the classic structuralist account of Fordism as a generalized set of production principles, historicized within the long cycles of historical capitalism. For Aglietta, although Fordist practices may have originated as specific policies within the auto industry, Fordism represents a societal-level shift in the relations of production and the reproduction of capital. In *A Theory of Capitalist Regulation*,¹⁵ he criticizes the dominant economic theories of capitalist equilibrium for being, among other things, ahistorical and abstract. Against this, Aglietta denies "that what exists does so automatically", necessitating a theory of social reproduction, and he

¹⁴Clarke, "What in the F—'s Name Is Fordism." This phrase does not appear in the version of this paper published in *Fordism and Flexibility: Divisions and Change*, but in an earlier (unpaginated) draft presented at the 1990 British Sociological Association Conference available at this web location: <https://scispace.com/pdf/what-in-the-f-s-name-is-fordism-2bwbsf1e0m.pdf>

¹⁵Michel Aglietta, *A Theory of Capitalist Regulation: The US Experience* (NLB, 1979).

claims that “when actual social systems are studied, historical experience confirms that *transformation means rupture, qualitative change*.”¹⁶ In other words, although Henry Ford's specific policies may have been solutions to the immediate problems in his own factories, their eventual widespread adoption does not indicate that they tapped into some transhistorical and abstract need for workers to receive higher wages. Rather, it indicates that these policies addressed a *historical* condition that may not have existed before, and may change again in time. Aglietta is inspired by "Marx's refusal to postulate an immutable essence underlying the variability of phenomena"¹⁷ and so he attempts to situate Fordism as a specific moment within the turbulent history of capitalist production.

Taking the US since 1870 as his spatial-temporal object, Aglietta elaborates a theory of capitalist regulation as an attempt to “formulate in general laws the way in which the determinant structure of a society is reproduced.”¹⁸ These are not transhistorical laws for society in general, but laws structuring a specific society. In the decades after the American civil war, they were oriented towards the development of heavy industry. Claiming that “the fundamental economic condition for industrial capitalism to develop is the formation of a growing agricultural surplus product and its realization in commodity form,”¹⁹ Aglietta highlights 19th century America's privileged situation. The incorporation of vast frontiers precipitated the development of rail infrastructure, irrigation, farming and mining. This ‘regime of extensive accumulation’ was funded and aided by the development of western urban centers and their growing financial powers, but it did not initiate a fundamental restructuring of the life of laborers. Labor was becoming more productive through intensification and mechanization, but the laborers' wages

¹⁶Aglietta, *Capitalist Regulation*, 12.

¹⁷Aglietta, *Capitalist Regulation*, 15.

¹⁸Aglietta, *Capitalist Regulation*, 13.

¹⁹Aglietta, *Capitalist Regulation*, 73.

were generally low, and they were often reliant on a significant amount of subsistence labor rather than commodity consumption at home.

In the first decades of the twentieth century, this regime fell into crisis. Aglietta positions the economic collapse of the 1930s as a result of overaccumulation and underconsumption. Specifically, the average worker was paid so little that they had to support themselves through a significant amount of informal labor, and therefore could not contribute to the valorization cycle by purchasing and consuming commodities. As capital entered a crisis in the 1930s, producers sought out new ways to maintain profitability. This would mean restructuring society to expand the domestic market and increase the class of consumers. As producers cast about for a model, Fordism, at this point a decade and a half old, stood out as a generalizable answer to changing economic conditions.

Fordism figures into Aglietta's account as an 'intensive regime' which "creates a new mode of life for the wage-earning class by establishing a logic that operates on the totality of time and space occupied or traversed by its individuals in daily life."²⁰ In other words, Fordism is characterized by an unprecedented expansion of capitalist interest and control over the lives of its waged workers beyond the shop floor. In order to stave off another crisis of overproduction and underconsumption, the daily lives of workers would have to be rebuilt on the foundation of commodity consumption. Aglietta terms this reorganization of social life a 'mode of consumption' which is oriented towards the renewal of the energy expended in labor processes and the maintenance of a continual supply of labor power.

As an articulation between production and consumption, Fordism comprised several specific innovations linked to more general social reconfigurations. On the production side,

²⁰Aglietta, *Capitalist Regulation*, 71.

Aglietta identifies the automated assembly line as Fordism's "characteristic labour process."²¹ The assembly line separated mental and manual labor to a previously unseen degree. Instead of managing the relationship between several related tasks, workers performed simplified rote tasks over and over: "the labour process [was] converted from a dense network of relationships between jobs, with intermediate products passing back and forth ... into a straightforward linear flow of the material under transformation."²² In Ford's factories, the production of a car was decomposed into 7,882 separate tasks, each of which was carefully documented and classified. The workers were similarly rationalized - not humans, but various assemblages of limbs: "The lightest jobs were again classified to discover how many of them required the use of full faculties, and we found that 670 could be filled by legless men, 2,637 by one-legged men, 2 by armless men, 715 by one-armed men, and 10 by blind men."²³

The mental component of labor was excised from the ambit of the general laborer and given to a growing class of managers and process engineers. Workers became fixed to clearly and rigidly defined roles that were "determined by the configuration of the machine system"²⁴ which was "planned in engineering departments which were themselves subject to the division of labor."²⁵ In this sense, the Fordist revolution went hand in hand with the managerial revolution described by Alfred Chandler.²⁶ As the scale and scope of production increased, and as jobs were broken up into smaller and smaller elements of an intricately designed whole, the "division between the labour of head and hand"²⁷ was deepened. Thought was no longer a part of the act of

²¹Aglietta, *Capitalist Regulation*, 117.

²²Aglietta, *Capitalist Regulation*, 118.

²³Henry Ford and Samuel Crowther, *My Life and Work* (Doubleday, Page & Company, 1922), 108.

²⁴Aglietta, *Capitalist Regulation*, 118.

²⁵Aglietta, *Capitalist Regulation*, 119.

²⁶Chandler, *The Visible Hand*.

²⁷Alfred Sohn-Rethel, *Intellectual and Manual Labour: A Critique of Epistemology*, trans. Martin Sohn-Rethel, Historical Materialism Book Series (1978; Brill Academic Publishers, 2020), 3.

labor itself, but something that had already been completed in the past and invested into the creation of a proscribed list of infinitely repeatable movements. The Fordist workplace began to look more and more like that described by Marx: living labor ruled by the dead labor that was embodied in the planning and manufacturing of the machine system.

This process of rationalization extended far beyond the shop floor. Along with the revolution in production, Fordism reconfigured the social and private lives of the workforce. Aglietta notes that “for the first time in history, Fordism created a norm of working-class consumption in which the general ownership of commodities governed the concrete practices of consumption.” Two commodities stand out as governing this norm: the automobile as a means of private transportation, and standardized housing. The adoption of these two major commodities transformed other patterns of household production. Households began to substitute the purchase of ready-made products like canned food and clothing for space-intensive home production, such as butchering and tailoring. Further, the nineteenth century pattern of roaming peddlers bringing such necessities as pots, pans or medicines, directly to the tenement buildings where workers lived, was replaced by workers taking their now-indispensable cars on trips to the grocer and hardware store.²⁸ All of these commodities transformed the urban spaces in which the working classes were renewed, and their relatively high cost “presupposed a vast socialization of finance, and correlatively a very strict control over workers’ resources and expenditures.”²⁹ However, Aglietta notes that the rise of certain family norms within a consuming class of workers was never quite general: Fordism also deepened the stratification of the working classes, which “forms part of the concrete conditions of reproducing the wage relation.”³⁰

²⁸Ruth Schwarz Cowen, *More Work for Mother: The Ironies of Household Technology from the Open Hearth to the Microwave* (Basic Books, 1983), 71-85.

²⁹Aglietta, *Capitalist Regulation*, 159.

³⁰Aglietta, *Capitalist Regulation*, 137.

Aglietta's account of Fordism as an historical articulation of means of production and consumption responding to the global needs of capital accumulation is a compelling one, but it fails on two counts, both related to his idealist conception of Fordism. First, Aglietta considers the intensive regime of accumulation to be an ideal type of capitalist society embodied by Fordist America, and against which other societies can be measured, rather than the outcome of a struggle between the concrete interests of specific classes and actors. Simon Clarke criticizes Aglietta, deeming his perspective structural-functionalist. Clarke claims that for the Regulationists, "in the relation between structure and struggle, it is the class struggle which takes the subordinate role, its significance constrained by the structural imperatives of the regime of accumulation."³¹ On the question of working-class struggle this charge is perhaps more applicable to Aglietta's followers than to his own work, which deals at length with working class organization. But the working classes are not merely structural features of a capitalist society. Workers are agents with the power to impose their demands on capital, and their organizations are not subordinate to the structure of capital accumulation, but constitutive of it.

This calls into question the regulationist framework of extensive vs intensive regimes of accumulation. Clarke criticizes the concept of a regime of accumulation, which first assumes a structural equilibrium and then attempts to explain it with recourse to a relatively stable configuration of institutional forms. For Clarke, the dynamic between production and consumption associated with Fordism are "best seen not as aspects of the regulation of a regime of accumulation, but as related aspects of the institutionalization of a particular balance of class forces in the immediate post-war period."³² In other words, the dynamic pioneered by Ford in the

³¹Clarke, Simon. 1988. "Overaccumulation, Class Struggle and the Regulation Approach." *Capital & Class*. Vol 12:3.

³²Clarke, "Overaccumulation," 24.

1910s and 1920s was established primarily through the concrete social and political struggles. The economic policy of Keynesianism was dictated by “the balance of class forces, institutionalized in the systems of industrial relations, electoral politics and the welfare state.”³³ In short, the regulation approach “overestimates the stability and duration of ‘Fordist’ modes of regulation” and “the extent to which Fordism can be considered a systematic ‘regime of accumulation’.”³⁴

The second shortcoming of Aglietta's work is his (non)treatment of imperialism, and his abstraction of US capitalism from the system of global capitalism. In the introduction to *A Theory of Capitalist Regulation*, Aglietta defines imperialism as a system “through which one state manages to influence a series of other states to adopt a set of rules that are favourable to the stability of a vast space of multilateral commodity relations guaranteeing the circulation of capital.”³⁵ After posing this anemic definition, he pushes it aside, claiming that “there could be little question of analysing in the present work the system of contemporary states and the transformation that the crises of the 20th century have produced in inter-state relations.”³⁶ But he is unable to escape the weakness of his conception of capitalist hegemony - it undermines his entire project. In his attempt to analytically isolate America from the Global South, he misapprehends the extent to which American industry both exacerbated and was dependent on an unequal global division of labor. The intensification of exploitation across state lines undergirds capitalist accumulation in Global North states. The insufficient definition of imperialism as management inflicted by one state on several with the goal of broadening the circulation of

³³Clarke, “Overaccumulation” 18.

³⁴Clarke, “Overaccumulation” 16.

³⁵Aglietta, *Capitalist Regulation*, 32.

³⁶Aglietta, *Capitalist Regulation*, 32.

capital prevents Aglietta from appreciating the global scope of capitalist regulation and reproduction.

In "Towards Global Fordism?" Alain Lipietz, another scholar from the regulation school, attempts to more adequately theorize the connection between the Fordist core and the periphery, but his analysis shows the same weaknesses as Aglietta's. Writing in 1982, he notes that

in three 'glorious' post-war decades, a crucial change was introduced by the system of capitalist accumulation that various French economists, following Gramsci, have baptized 'Fordism.' A 'centre', displaying a cohesive internal logic at the level of blocs (North America, the EEC) if not individual countries, came to stand against a 'periphery' which, whatever the internal dynamic of its component parts, was ever more disconnected from this 'autocentred' logic.³⁷

However Lipietz immediately questions whether we can even "speak of a world-wide extension of Fordism," noting that during the late 1960s and early 1970s, the economic periphery experienced uninterrupted growth while the economic core went into a production crisis. Indeed, the industrialization of the periphery was not a slow march following in the footsteps of the core, but was its own specific development, conditioned by and conditioning the dynamics of accumulation in the United States and Europe. By the time 'Fordism' reached the Third World, it had devolved into another one of its non-ideal, mutated forms. Lipietz describes a Global South Fordism stripped of its impetus to consumption as "sub-Fordism" and "'Bloody' Taylorization."³⁸

Although Lipietz's work is rigorous and empirically rich, it is undergirded by a flawed methodology. He first maintains Aglietta's idealist assumption that the Fordism of the US can be analytically extracted from its own contemporary global context. He then looks at the structures of accumulation in the Global South decades later and compares them to this ideal type. The implicit question is not 'how did the structure of accumulation in the South develop?' but 'why

³⁷Alain Lipietz, "Towards Global Fordism?," *New Left Review*, no. 132 (1982): 33–47.

³⁸Lipietz, "Towards Global Fordism?" 41.

did Fordism fail to be exported as-is?' It completely sidesteps the question of how production in the South is contributed to the development of Fordism in the core.

A more appropriate approach to the global scope of Fordism is to trace the specific ways in which the automotive industry created new relations of production and consumption in peripheral countries, and how these relations contributed to the accumulation of capital in the core countries. A useful framework for such an analysis is the commodity chain - the global network of extraction and production processes that transform raw materials into finished goods. As an raw essential material unavailable in the US and Europe, rubber and its conditions of production expose the fundamentally global nature of the production of automobiles. Cars could not be produced if rubber was not available - and their production would not be profitable if it was too expensive. In the drive to procure sufficient quantities of cheap rubber, European and American capital pulled the rubber producing regions in the Amazon, Africa and Southeast Asia into their sphere of influence, but on deeply unequal terms.

Several factors contributed to the underdevelopment of the rubber regions: the demands of Northern industry for cheap rubber, Colonial governments with no interest in the development of stable and autonomous economies in the Global South, and unscrupulous local actors happy to exploit their countrymen for a marginal share in the profits of capital. Each of these factors interacted with the same categories that were so important to Henry Ford: the possibilities of rationalizing processes of production and the substitution of commodity consumption for subsistence labor. These two elements of re/production were manipulated in different ways than in the US Midwest, but just as in the North, they were unified as part of a "logic that operates on the totality of time and space occupied or traversed by its individuals in daily life."³⁹ To sideline

³⁹Aglietta, *Capitalist Regulation*, 71.

the exploitation of rubber workers, and the exclusion of Global South industry from the profits of the automotive industry is to misrecognize Fordism as a category of the capitalist world-system.

If we have questioned the definition of Fordism as a systematic regime of accumulation, and its validity as a mode of regulation divorced from both state action and the global contours of capital, we might reasonably ask what use the concept of Fordism provides, and what value we can gain from the regulation school's treatment of it. Two of Aglietta's core interventions - Fordism as a specific qualitative change in the articulation of production and consumption, and the increased stratification of the wage-earning class - are crucial, but they must be reformulated as questions. We are interested in asking what concrete struggles led to the particular relationship between production and consumption in the early automotive industry, as well as interrogating the limits of this relationship. Furthermore, we cannot define Fordism so narrowly as to exclude the conditions of production in the Global South. The production of rubber in the Amazon, Africa and Southeast Asia were not external to the automotive industry, but internal to it. With this in mind, we will investigate the articulation between production and consumption in the rubber industry, and the stratification of the rubber workers. We will especially pay attention to the development of commodity consumption and attempts at the rationalization of rubber gathering. In the following sections, we will look more closely at these phenomena both in the early Michigan automotive industry, and in the rubber industry abroad.

A Provincial Study: Fordism in Michigan

As the birthplace of Fordism, however defined, industrial Michigan bears the imprint of Fordism's early development. Since the Second World War, American labor historians have been arguing over these dynamics. Marxist historians in the US emphasized the labor process itself as

an essential component of labor history. Harry Braverman's *Labor and Monopoly Capital: The Degradation of Work in the Twentieth Century* centered the deskilling of the workforce as a method of labor control. Nelson Lichtenstein writes in the introduction to *On the Line: Essays in the History of Auto Work* that "Braverman broke new ground by alerting historians and sociologists to the social context within which the technology of production was embedded."⁴⁰

Braverman's view was oversimplified, however, and labor historians later debated "the extent to which craft skills were actually eroded in mass production industry."⁴¹ Rather, the labor market was bifurcated between low-paid and low-skilled workers and highly paid engineers and managers. The partial deskilling of the workforce was a point of tension between labor and capital. At times capitalists strategically degraded the skill of workers in order to disempower them, while the unions resisted this degradation.

It is often repeated that unions in late 19th and early 20th century Michigan were weak and listless - supposedly this contributed to the rise of Fordism as a new method of labor exploitation. However, this claim is challenged by Thomas Klug, in an essay in *On the Line*. As the lumber industry declined at the close of the century, economic activity in the state shifted towards the growing cities. In the 1890s, Detroit was the stove producing capital of the world, with a significant presence of other types of metalwork that would be crucial to the development of the auto industry. Most of the organizations representing workers were craft unions, and they admitted only skilled labor to their ranks. However, the assertion that these unions were powerless is misguided. Klug notes that "although only a few employers of iron molders formally agreed to the exclusive employment of union men, many foundries were de facto closed

⁴⁰Nelson Lichtenstein and Stephen Meyer III, eds., *On the Line: Essays in the History of Auto Work* (University of Illinois Press, 1988), 5.

⁴¹Lichtenstein, "Introduction," 5.

shops.”⁴² The reason for this is the Iron Molders Union’s strategy of organizing the “best molders” so that any employer wanting good labor had to go through the union.

The craft unions in fact posed enough of a threat to employers that in 1902, “metal trade employers organized the Employers’ Association of Detroit (EAD).”⁴³ They felt that “the brief accommodation with labor unions in the late 1890s and early 1900s had gained them few benefits Rather, strikes became more frequent, and unions gained additional recruits.”⁴⁴ The unions had developed significant leverage in eastern Michigan and prevented employers from undermining labor rights to increase productivity and profitability. As a unified bloc, the EAD was able to enforce what they considered to be their right to pay workers based on the quality of their individual work, and to hire from outside the local labor pool.

Detroit employers were also troubled by what they saw as a decrease in control over the labor process itself. As the unions had a monopoly on skilled labor, which was crucial to the operation of heavy manufacturing, they were able to exercise significant influence on the operations of firms. John J. Whirl, a member of the EAD, said that “in some of our industries the proprietors were merely the financial agents to purchase the supplies, pay the wages, and sell the manufactured products ... As to the operations of the factory, they had no more real control than if they were in no way connected with it.”⁴⁵ The attempt by employers to band together and undermine the craft unions was joined with an understanding that to truly gain control of their businesses, they would need to reduce their reliance on a small pool of skilled labor: an understanding that would animate the coming Fordist revolution.

⁴²Thomas Klug, “Employers’ Strategies in the Detroit Labor Market, 1900-1929,” in *On The Line: Essays in the History of Auto Work*, ed. Nelson Lichtenstein and Stephen Meyer (University of Illinois Press, 1989), 44.

⁴³Klug, “Employers’ Strategies,” 48.

⁴⁴Klug, “Employers’ Strategies,” 48.

⁴⁵Klug, “Employers’ Strategies,” 49.

And it was a revolution. The automobile was not invented in Michigan – in fact the first two viable gasoline powered cars were built in Massachusetts and Indiana in 1892 and 1894.⁴⁶ But Michigan arrived early on the automotive scene, propelled by easy access to machining, raw materials, and capital: in 1896 Ransom E. Olds of Lansing and Detroit's Charles B. King and Henry Ford demonstrated viable motor powered carriages. By the early twentieth century, there were dozens of auto companies in Michigan, but they faced significant issues.

Initially, auto production was a craft industry: each car was handmade by skilled craftsmen. Due to the competitive market in Michigan, it was difficult for employers to keep workers from leaving for greener pastures at another company. The turnover rate was over 200% in Detroit plants, reaching a high of 382% at Ford's Highland Park plant, and the costs of training new workers was estimated to be between \$35 and \$100.⁴⁷ Meanwhile, the cars were also expensive, and so there was a limited market for the product being produced. Henry Ford's management approach targeted both of these problems, and ultimately transformed the industry.

In 1908, Ford introduced the Model T, a mass produced car with standardized parts that required less skill to make and cost far less than competitor's cars, contributing to a rise in sales. In 1913 he introduced his version of the automated assembly line, further standardizing production and reducing costs. And in 1914, he initiated the Five Dollar Day program, a wage increase that solved the problems of employee turnover and bloated training costs. In order to compete, rival firms had to adopt and even advance Fordist innovations. In the 1920s, General Motors introduced the yearly model change to stimulate consumer demand even further.

⁴⁶Willis F. Dunbar, *Michigan: A History of the Wolverine State* (Eerdman, 1980), 552.

⁴⁷Klug, "Employers' Strategies," 54.

The challenges that Fordism introduced to the labor market and to the shop floor have been of particular interest to labor historians working in the Marxist tradition. From the early suppression of union activity by auto companies to the 1936 sit-down strike in Flint and the rise of the most militant unions in American history, the Fordist factory is a towering edifice in American labor historiography. But a full accounting of the social change wrought by Fordism requires a labor history that is critical of the notion of labor itself.

Liberal and Marxist historians in the twentieth century disagreed about the political nature of their work and the goal of the labor movement, but there is a fundamental similarity in their analysis. The key category for both traditions is labor, but while labor historians interrogated its relations with other social forms such as capital or law, they were less critical of the category of labor itself. Labor and by extension laborers are taken to be self-evidently coherent classifications: generally wage-laborers, generally men, generally working for someone else.

In 1971, Thomas Krueger, in “American Labor Historiography,” narrates a partial break with the shared paradigm of the liberal and Marxist historiography, but at the time of his writing that break was still incomplete. He is concerned with a shift away from the traditional conception of labor history, citing Hyman Berman’s opinion that labor history “is more than the rise or fall or the failures and successes of organizations,” and David Montgomery’s call to research “the cultural and intellectual life of the working classes.”⁴⁸ This new direction follows the work of the British Marxists in expanding the category of the working classes to include a greater sample of

⁴⁸Berman and Montgomery quoted in Thomas A. Krueger, “American Labor Historiography, Old and New,” *Journal of Social History* 4, no. 3 (1971): 281-2.

the relations of working class life, including community support of labor against capital during strikes, the influence of religion, and the permeability of class categories.

Krueger notes that in 1971, this development was still immature, and had to “solve some of its intellectual problems.”⁴⁹ His framing of the task ahead, however, demonstrates just how partial the break with old labor history still was. He lists ten “imposing questions” that need to be answered by new labor historians, and while they demonstrate an expansive view of labor *history*, they hew to a decidedly traditional view of the category of labor itself. One question is representative: “What happened to blacklisted workers, to men over 40 who got fired, to families whose head was mangled or killed on the job.”⁵⁰ Implicit in this question is the orientation of a nuclear family around labor and the elevation of specifically male waged labor.

In the 1970s, a significant challenge was posed to this traditional notion of labor, and to both Marxist and classical labor historiography, by feminist organizations and historians. Prominent among these was the Wages for Housework campaign, and the challenge they posed changed the way that labor history in the US could be written. Writers and activists in this tradition questioned the notion that only waged labor contributed to the economy and took traditional Marxism to task for accepting the capitalist devaluation of feminized work.

Sylvia Federici, in the 1975 essay “Counterplanning from the Kitchen” lays out the feminist challenge to a wage-centric notion of labor posed by Wages for Housework:

Beginning with ourselves as women we realize that working for capital does not necessarily produce a paycheck and does not begin and end at the factory gates.... Housework, in fact, is much more than housecleaning. It is servicing the wagedworkers physically, emotionally, and sexually and getting them ready to work day after day for the wage.... This means that behind every factory, every school, every office, and every mine

⁴⁹Krueger, “American Labor,” 283.

⁵⁰Krueger, “American Labor,” 284.

there is the hidden work of millions of women, who consume their life reproducing those who work in those factories, schools, offices, and mines.⁵¹

The family life of workers, in other words, is not a realm separate and apart from their labor but is constitutive of it. The unpaid work of women (and children) forms the base on which the institution of waged labor stands. Implicit in Federici's quote (and explicit in her other work) is the accusation that labor history and labor activism, by ignoring the work of women, reproduces capital's own devaluation of gendered and non-waged work.

The point of this intervention is not to merely add a new type of labor to an old conception: the new category is not labor plus gender. A labor history that merely traces struggles and changes in the structure of housework would do little more than transplant the old framework of analysis onto a new subject. Likewise, a treatment of gender as a variable that affects only women would be counterproductive. Rather, the task for feminist labor history is to interrogate the very process of categorization. Elizabeth Faue, introducing a volume of essays on gender and labor history, claims that "labor historians' use of collectivities as units of analysis – whether classes, communities, trades, or families – masked both intraclass division and an understanding of conflicting interests that was at times one-dimensional."⁵² To remedy this, we must examine not how the category of gender *acts on* the category of labor, but how it permeates and constitutes it.

The work of Marxists in theoretically elaborating the role of the household in capitalist reproduction is useful here. Marxist feminists have critically investigated the categories of the family and the household, and how they have been manipulated to serve the interests of capital. Within the household, labor is reproduced through a combination of inputs, chief among them

⁵¹Silvia Federici, *Patriarchy of the Wage: Notes on Marx, Gender, and Feminism* (PM Press, 2021), 11-12.

⁵²Elizabeth Faue, "Gender and the Reconstruction of Labor History: An Introduction," *Labor History* 34, nos. 2–3 (1993): 171.

gendered and devalued domestic labor. The structure of labor reproduction is not constant, but changes in response to different circumstances/locations. In the work of Social Reproduction Theorists, devalued domestic labor takes center stage as the condition of the reproduction of wage laborers, and so they make a general distinction between productive labor (generally that which is done for a wage) and unproductive/reproductive labor (that which is concerned with providing for the base needs of life, on either a waged or unwaged basis). As Nancy Fraser puts it, the unwaged work of (usually) women supports the renewal of the life of the waged worker and forms “one of [the] background conditions of possibility” of capitalist accumulation.⁵³

The household, however, serves more purposes than just the reproduction of wage-laborers; they are also the ultimate sites of commodity consumption. Kirstin Munro writes that “[w]orking class households are compelled to engage in household production not only to survive, but so their members can appear continually on the market as renewed labour-power in order to continually buy more commodities.”⁵⁴ Because households in capitalist societies are severed from traditional means of subsistence, they are forced to resort to commodity consumption in the maintenance of their lives. This means that “the household and its production processes are implicated in the reproduction of capitalist society”⁵⁵ - households are not merely the objects of a devastating transformation effected by capitalist society. Through commodity consumption, they continually reproduce the necessary conditions for the existence of capitalism.

⁵³Nancy Fraser, “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism,” in *Social Reproduction Theory: Remapping Class, Recentering Oppression*, ed. Tithi Bhattacharya (Pluto Press, 2017), 23.

⁵⁴Kirstin Munro, “THE CAPITALIST FORM OF HOUSEHOLD PRODUCTION: SUBSIDY, ANACHRONISM, OR SOMETHING ELSE,” *Etica & Politica / Ethics & Politics* 27, no. 1 (2025). 266.

⁵⁵Kirstin Munro, “‘Social Reproduction Theory,’ Social Reproduction, and Household Production,” *Science & Society* 83, no. 4 (2019). 454.

This is not an assignment of blame, but a diagnosis of the mutually conditioning nature of the relationship between households and the capitalist society in which they are embedded.

Marxist Feminist perspectives on the household can grant us insight into the management of labor in Detroit around the turn of the century. Prior to 1914, when Henry Ford instituted the famous Five Dollar Day in an attempt to reduce employee turnover, it was uncommon for Michigan families to live on the wage of a single earner. Working-class families cobbled together livings from various sources, fighting both to make higher wages and to carve out more time to do necessary subsistence labor. In “The Detroit Story: the crucible of Fordism,” Kathleen Stanley and Joan Smith examine the extent to which working-class families were transformed by the family-wage in Detroit from roughly 1890 through the middle of the twentieth century. They build on the category of the household as an income-pooling unit from world-systems analysis and take the family as a site of wage-contestation. Working-class households piece together subsistence from various sources: in addition the consumption of commodities purchased with a wage, they rely on the unpaid subsistence labor of women and children, charity or community assistance, income from boarders, informal commodity production, and many other sources. The need to depend on these extra-wage sources of income would be alleviated with a family-wage.

Smith and Stanley point out that in the 1890s, wages in the Detroit area were much lower than the national average. This was partly due to the depressed wages of immigrants who made up close to 70% of the city, and due to the relatively large percentage of the workforce that were women. While noting that there was much variation among family incomes, they cite that “The [average] laborer whose budget was reported by the Michigan Bureau of Labor, for example,

faced an annual shortfall of \$86, or 27 percent of his total income.”⁵⁶ Over a third of the city’s workforce was made up of “semiskilled, unskilled, and service workers” who earned considerably less than this average.

In addition to the type of labor performed, a major factor in the ability of Detroit workers in 1890 to survive on a wage was family composition, which correlated strongly with race. Polish immigrants generally had the largest households of any ethnic group, much greater than American-born whites. While most of the households across all ethnic groups were organized as nuclear families, Black families were far more likely to either not have children, or to have only one parent. Stanley and Smith note that while smaller families are generally considered to be more sustainable in times of depression (such as the 1890s), the significantly larger-than-average Polish households were able to leverage their family composition into more effective “patterns of subsistence.”⁵⁷

It appears that “Polish families brought with them skills associated with their recent past as peasants, which could be put to use outside the wage nexus.”⁵⁸ While it was uncommon for married Polish women to work outside the home for a wage, they were more likely than other groups to maintain livestock and vegetable gardens. When cash flow was an issue, Polish children were sent to work at higher rates than children of other families. The various forms of subsistence labor and multiple sources of wage income combined to insulate Polish families from both economic downturns and a reliance on a single wage-earner.

⁵⁶Kathleen Stanley and Joan Smith, “The Detroit Story: The Crucible of Fordism,” in *Creating and Transforming Households: The Constraints of the World-Economy*, ed. Joan Smith and Immanuel Wallerstein, Studies in Modern Capitalism (Cambridge University Press, 1992), 36.

⁵⁷Stanley and Smith, “The Detroit Story,” 37.

⁵⁸Stanley and Smith, “The Detroit Story,” 38.

Other ethnic groups in the 1890s had their own strategies for survival that combined wage and non-wage incomes. Black families in Detroit generally had fewer dependents, which lowered the cost of living, but also meant that they could not send children to work. As a result, “married Black women were twice as likely to work for wages as other women”⁵⁹ and over a third of all Black women had waged jobs. Additionally, by 1900, Black families had the highest proportion of boarders in the city – non-family members who earned their keep either through rent payments or through contributing subsistence labor to the family.

American born whites also depended on boarders, although to a lesser extent than other ethnic groups. They generally had children but pulled their children out of school to work at a lower rate than other ethnic groups. Stanley and Smith note that “this was possible in part because of the higher wages received by native White men, 72 percent of whom were either professional or skilled workers.”⁶⁰ These families rarely sent married women into the workforce, although occasionally unmarried white women did work outside of the home.

After the development of the Five-Dollar Day in 1914, household patterns in Detroit changed, but not necessarily in the direction of single-income nuclear families. Stanley and Smith note that in the decades following Ford’s family-wage campaign, “even in Detroit, this presumed bastion of Fordism, the family wage was never a reality for the majority of wage-earners.” Except for a small elite strata of the working classes, “the male wage was always supplemented by the wages of children and sometimes of wives, as well as by considerable amounts of subsistence labor.”⁶¹ This picture looks much the same as pre-Fordist Detroit, so what changed? Was the social transformation of Fordism as radical as its labor transformation?

⁵⁹Stanley and Smith, “The Detroit Story,” 39.

⁶⁰Stanley and Smith, “The Detroit Story,” 38.

⁶¹Stanley and Smith, “The Detroit Story,” 34.

In short, yes, but to understand exactly how we need to look more closely at patterns of race. Between 1900 and 1940, the population of Michigan more than doubled, from 2.4 million to 5.3 million. During this same period, the Black population of the state increased by a factor of more than *thirteen*, from just over 15,000 to more than 208, 000.⁶² Most of this increase was driven by Black migrants from the southern states, looking for better working conditions and wages in the industrial Midwest.

To an extent they were able to find them, but they did not enter the Michigan workforce on equal footing with whites. According to the 1930 census, Black men were significantly overrepresented in low paid manufacturing jobs, while their white counterparts were more often employed either as shop foremen or skilled machinists.⁶³ The cities themselves mirrored this, with white workers more often owning free standing homes in newly developed neighborhoods, and Black workers and poor whites confined to the slums. In other words, the social significance of Fordism is defined as much by who it excluded from prosperity as by who it included.

Disparities in income are compensated for in many ways, and as Stanley and Smith demonstrate, this is an essential element in labor history. It is not enough to note that Black laborers generally earned less than whites – we must look at the household forms that propped up this disparity and that allowed low-income groups to survive despite their condition of degraded labor. Stanley and Smith noted that Black families were much more likely to rely on boarders to make ends meet, and Black women were much more likely to work outside the home. Even within the “white” ethnic groups, there are various forms of family composition that contributed to the types of labor performed. Polish women laboring informally in vegetable gardens or caring

⁶²US Census Bureau. *Michigan*. 1940.

⁶³US Census Bureau. *Michigan*. 1930.

for chickens are as much a part of labor history as American born unmarried white women working outside the home for a wage. And both are inextricable from the histories of the generally male, generally white worker that take up so much space in liberal and Marxist labor histories.

Global Complications: the Rubber Industry

Fordism did not develop in isolation, nor exclusively from local dynamics. Just as the existence of Fordist production methods was dependent on racial and gendered divisions in the local labor market, it was also built on a global division of labor, whereby the more brutal exploitation of workers in peripheral states propped up the productive relations in the core. Wallerstein refers to this global dynamic as the “axial division of labor,” a dynamic which ensures a “constant flow of surplus-value from the producers of peripheral products to the producers of core-like products.”⁶⁴

The nature of this global dynamic between the capitalist core and periphery was characterized by Andre Gunder Frank as the development of underdevelopment.⁶⁵ In his account, the supposed economic backwardness of the Global South is not a failure to keep pace with the capitalist core, but rather “is the result of its centuries-long participation in the process of world capitalist development.”⁶⁶ In other words, the underdevelopment of the Global South is a condition positively created by the dynamics of global capitalism. Samir Amin’s *Accumulation on a World Scale* deals with this concept as well, attempting to counter the prevailing stagist narrative of underdevelopment which isolates the poverty of the periphery from world history

⁶⁴Immanuel Wallerstein, *World-Systems Analysis: An Introduction* (Duke University Press, 2004), 60.

⁶⁵Andre Gunder Frank, “The Development of Underdevelopment,” *Monthly Review* 18, no. 4 (1966): 17–31.

⁶⁶Frank, “The Development of Underdevelopment,” 21.

and assumes that “laissez-faire was certain to develop the colonies just as it had developed the industrialized metropolitan countries.”⁶⁷ Amin insists that it is the capitalist core’s economic growth itself that creates the impoverished condition of the periphery. He claims that “as economic growth proceeds, none of these features by which the structure of the periphery is distinguished lessens; on the contrary, each increases.... In the periphery growth is *not* development, for its effect is to disarticulate.”⁶⁸

Economic disarticulation in the periphery is central to Amin’s analysis. In the core, national economies are largely integrated: different sectors “carry out substantial exchanges between themselves, which may be called “interindustrial” or “intersectoral” exchanges.”⁶⁹ The outputs of extractive industries within developed economies directly support domestic manufacturing and agriculture. For so-called “underdeveloped” economies, internal trade between industrial sectors is stifled: “The different kinds of mineral wealth exploited by these great concerns - metals, oil, etc. - are not destined to supply domestic industries on the spot, but are exported in order to supply complex industrial groups in the advanced countries.”⁷⁰ Whereas in the core, rationalization and growth in one industry is quickly generalized to others, in the periphery industrial growth has minimal effect on the domestic economy as a whole, transferring the benefits to its industrial trading partners.

This dynamic can be traced along what are called commodity chains: the global networks of labor and raw-material inputs that feed into the production of particular industrial and consumer goods. The term commodity chain was coined in a 1977 paper by Wallerstein and Hopkins, who propose it as an analytical tool to combat a “developmentalist” perspective in

⁶⁷Samir Amin, *Accumulation on a World Scale* (Monthly Review Press, 1971), 13.

⁶⁸Amin, *Accumulation*, 18.

⁶⁹Amin, *Accumulation*, 16.

⁷⁰Amin, *Accumulation*, 16.

which “the world is said to consist of a number of related but basically autonomous ‘societies’ ... each moving upward along an essentially similar path of development.”⁷¹ Criticizing an analytical nationalism which attempts to abstract autonomous societies from each other, Wallerstein and Hopkins insist on analyzing the world-system *in toto*: “a definite ‘world’, a spatio-temporal whole, whose spatial scope is coextensive with the elementary division of labor among its constituent regions or parts and whose temporal scope extends for as long as the elementary division of labor continually reproduces the ‘world’ as a social whole.”⁷² Within this global social body, commodity chains form a sort of vascular system, transporting raw materials and ossified labor inputs from their sites of production, combining them, and bringing them to market where they can be valorized and turned into profit.

Commodity chains bring local and global production processes into contact. As commodity production expanded over the course of capitalism’s global history, the need for vast amounts of raw materials to be transported across large distances increased. Bunker and Ciccantell claim that the “solutions to the raw materials problem require the coordination of multiple physical and social processes across geopolitical and physical space with domestic relations between firms, sectors, the state, labor, and new technologies.”⁷³ As a sort of vascular system for raw and processed materials within the capitalist world-system, commodity chains are a primary axis of contestation for actors across ‘geopolitical and physical space.’ The state acts on commodity chains through trade agreements, or through nationalizing or privatizing reserves of raw materials. Labor asserts its power to strike or withhold itself at various points along the

⁷¹Hopkins, Terence K., Immanuel Wallerstein. 1977. “Patterns of Development of the Modern World-System.” *Review (Fernand Braudel Center)*, 111.

⁷²Hopkins and Wallerstein, “Patterns of Development,” 112.

⁷³Stephen G. Bunker and Paul S. Ciccantell, *Globalization and the Race for Resources* (Johns Hopkins University Press, 2005), 19.

chain - labor inputs being the stuff that connects various links. And finally the firms, capitalists themselves, act to either subsume more and more of the production processes that feed into the final consumable as in Fordist vertical integration, or they outsource moments in the production process to locales with a cheaper labor supply as we see in the trends of post-Fordist globalization.

Commodity chains are a useful concept for countering perspectives which treat capitalist accumulation as a local phenomenon. Analyses, such as Aglietta's, that treat capitalist production in core regions as a self-contained whole fail to account for how productive relations in the core are propped up by quite different productive relations in the periphery. Smith, Ciccantell, and Sowers note that this problematic has been taken up at length by world-systems analysts who examine the racialized nature of labor in "resource frontiers and extractive peripheries."⁷⁴ The effect of these "extractive peripheries on existing population, and especially indigenous populations, are almost universally negative, engendering often strong and violent opposition from indigenous groups."⁷⁵ The remainder of this section will examine production and reproduction in this context along the Fordist commodity chain.

For early automobile producers in Michigan, many of a significant portion of the commodity chain was entirely internal to the US. Many of the raw materials needed for the big three automakers to produce automobiles were readily available within the United States. Michigan's vast lumber and mineral resources were brought to Detroit by a robust rail system, and Ford even owned "mines in Kentucky and West Virginia, which gave him control over every

⁷⁴David A. Smith et al., "Reconstructing Commodity Chain Analysis as World-Systems Analysis," in *World-Systems Analysis at a Critical Juncture*, ed. Corey R. Payne et al. (Routledge, 2023), 62.

⁷⁵Smith et al, "Reconstructing Commodity Chain Analysis," 62.

natural resource needed to make a car - save rubber.”⁷⁶ To acquire the rubber necessary for the production of automobiles, Ford and his competitors had to look outside the US - at first to the Amazon and the Congo, and then later to Malaya and Sumatra in Southeast Asia.

Rubber has been known and used by Mesoamerican peoples for thousands of years - it was used to make bottles, balls or waterproof shoes. Despite this, in the 19th century the rubber industry in Europe and the US mainly provided for products with limited industrial practicality until the development of the vulcanization process in the late 1830s. Vulcanization allowed for the creation of stable and hardened rubber industrial products, and by 1845 carriage wheels were equipped with solid rubber tires.⁷⁷ The demand generated by tire production, along with rubber insulation for electrical wires, drove the rubber industry to new heights. In the US, the rubber industry was centered in Ohio - by 1900 Akron was the home to Goodrich, Goodyear, and Firestone. Business was good and all three were multi-million dollar companies by 1910. Firestone was especially successful, thanks to exclusive contracts with Ford to provide tires to the Model N in 1906 and to the famous Model T in 1908.⁷⁸ Fortunes were made in the manufacture and sale of tires, but down the commodity chain the story was quite different.

Large scale extraction of rubber began in Brazil and central Africa in the 1800s, where latex producing plants grew natively. Bunker and Ciccantell claim that since “the Amazon provided the technically most useful and abundant kinds of rubber in the world, states and firms in the rapidly industrializing, trade-dominant nations of Europe and America were particularly interested and active in the Amazon.”⁷⁹ As in the Congo, initial production consisted of wild-

⁷⁶Greg Grandin, “Empire’s Ruins: Detroit to the Amazon,” in *Imperial Debris: On Ruins and Ruination*, ed. Ann Laura Stoler (Duke University Press, 2013), 118.

⁷⁷Austin Coates, *The Commerce in Rubber: The First 250 Years* (Oxford University Press, 1988), 49.

⁷⁸Coates, *Commerce in Rubber*, 134.

⁷⁹Bunker and Ciccantell, *Globalization*, 39.

collection by contract laborers. Rubber plants were quickly exported to tropical regions around the globe such as southeast Asia, although there was little commercial production in Asia until the collapse of the rubber market in the Congo and the Amazon in 1913.⁸⁰ Although plantations would dominate the global rubber production by the second decade of the twentieth century, for the first several decades of the auto industry there were few rationalized rubber plantations, and rubber was sourced from wild collectors in the Amazon and the Congo. By the end of the nineteenth century, the Amazon was supplying the world market with 15,000 tons of rubber annually.

The working conditions in rubber fields of the Amazon were a world away from those in the US. In the late 1870s a severe drought struck Northeastern Brazil, destroying the means of subsistence for a large portion of the population. Many left their homes and found work in the growing Brazilian rubber industry, but under the harshest of conditions. Forced to pay for their own provisions on the way to the work locations, rubber tappers started working in severe debt. Further, the workers were cut off from their previous communities, and so to survive they were forced to rely on local rubber agents. Their debt was perpetuated by the sale of “the necessary working tools and basic goods such as foodstuff, which were forcibly acquired at the rubber patron’s storehouse or station.”⁸¹ The rubber companies held this debt on account against future rubber yields, but “every time [the worker’s] rubber (his credit) was collected, it was found that his debt had mounted higher than his earnings.”⁸² The rubber agents and traders ensured that the prices of necessary provisions were higher than the piece-wages for rubber, creating a vicious

⁸⁰Coates, *Commerce in Rubber*, 115.

⁸¹Louise Cardoso de Mello and Sven Melkebeke, *From the Amazon to the Congo Valley: A Comparative Study on the Violent Commodification of Labour During the Rubber Boom (1870s–1910s)*, Palgrave Studies in Economic History (Palgrave Macmillan, 2019), 148.

⁸²Coates, *Commerce in Rubber*, 94.

cycle of debt peonage. It was this debt relation, rather than a wage, which ensured the continual supply of labor for the rubber industry.

The labor supply in the Congo was ensured through similar means, but this time with a tax burden rather than a debt burden. In a comparative study of Amazonian and Congolese rubber extraction, de Mello and Van Melkebeke note that from the 1890s on, an “in kind rubber tax ... proved to be the solution to increase revenue.” They go on to say that “this huge increase in exports received by in-kind taxation would result in severe and violent labour mobilisations and would cause devastation throughout the Congolese countryside.”⁸³ Whereas in the Amazon, workers migrated to the rubber regions due to harsh economic and environmental conditions back home, in the Congo the state and private firms sent expeditions into the countryside to capture and enslave Indigenous men. “Expeditions often ended up in plunder, destruction of villages and capturing of people: initially, to collect ivory (by the Congo Free State) and subsequently, to collect rubber (by the Congo Free State and private sector).”⁸⁴ These workers were then forced to collect and deliver “six to eight kilograms per person every two weeks” and “when they did receive some sort of remuneration, this was usually in-kind (salt, knives, beads, textiles, etc.).”⁸⁵

Unlike in the US Midwest, few provisions were made for generational reproduction. It is true that the automotive industry in Michigan employed large amounts of immigrant labor, but if they conformed to Ford's social expectations then allowances were made to support their families. The Five Dollar Day was explicitly an encouragement to generational reproduction. In the Amazon and Congo, however, existing communities and their social relations were violently

⁸³de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 143.

⁸⁴de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 147.

⁸⁵de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 152-153.

disrupted by rubber barons and state actors who saw no need to recreate conditions for the renewal of laborers. Rather indigenous and migrant populations were almost exclusively an external pool from which new labor could be drawn. The pay given to rubber workers (if it can be called pay) was insufficient to provide for the renewal of the conditions of labor on an open labor market, and so the gap was filled by coercion, and by the workers' informal reproductive labor. Furthermore, although workers along the commodity chain were increasingly commodity consumers, as well as producers, in Michigan this took the form of household appliances purchased with an earned wage. In the rubber fields these commodities were obtained from closed company shops, and in both the Amazon and the Congo, roving rubber tappers lived not in their own houses, but in temporary shelters in the rubber fields.⁸⁶ When debt and tax burdens were insufficient to compel labor in these conditions, violence was used, and when violence was insufficient workers were killed, and new workers brought in.

We see that in the wild rubber fields, gendered labor dynamics differed significantly from those in industrial Michigan. Many Michigan women performed unpaid reproductive labor to support and supplement their husband's 'family wages' and in the poorer communities they took on piecework or service work in addition to this unpaid care work. In the core, this gendered dynamic was structured in part by the housing policies of the industries, but in the periphery the gendered division of labor manifested quite differently. Maria Mies' analysis of women and colonialism helps us contextualize the dynamics of gendered labor in the periphery. Writing about the 18th and 19th centuries, she notes that where unfree labor was secured through slaving raids and trade, "slave women were actively discouraged from bearing children or forming

⁸⁶de Mello and Melkebeke, "From the Amazon to the Congo Valley," 153.

families.”⁸⁷ As long as the slave trade was thriving and the problem of reproducing the labor force was resolved through external means, enslaved women were expected to perform hard labor alongside men. When the Atlantic slave trade was abolished, this dynamic shifted, and enslaved women were forced to bear and raise children - future workers. Mies notes that colonial administrators enacted “incentives in the slave codes of the late eighteenth and nineteenth centuries to encourage local breeding of slaves by slave women on the plantations.”⁸⁸

How did these dynamics play out in the twentieth century in the rubber fields? de Mello and Melkebeke seem to indicate that the 19th century practices of slaving raids in the Congo, in which “people were captured and enslaved, usually men, but also women, who were incorporated into the raiders’ harems,” were continued by rubber producers in the 20th century.⁸⁹ However, women also occasionally worked in the rubber fields. As a result of this system, the Congo witnessed a significant demographic decline, with “contemporary observers and present-day scholars [agreeing] on rural depopulation and decimation of the Congolese,” along with “numerous cases of depopulated villages”⁹⁰ - homes standing empty save for ghosts. The situation in the Amazon was analogous: “In general terms, the demographic post-boom scenario is marked by the emptying of villages or whole regions, being caused either by workforce recruitment, the resulting unsustainable livelihood, death by diseases, ethnocide, demographic replacement and miscegenation or encroachment.”⁹¹

The wild rubber industry was quite volatile, in large part because of the decentralized nature of the production system. Rubber in Brazil was not produced on plantations, but was

⁸⁷Maria Mies, *Patriarchy and Accumulation on a World Scale: Women in the International Division of Labour* (Zed Books, 1986), 91.

⁸⁸Mies, *Patriarchy and Accumulation*, 91.

⁸⁹de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 145-146.

⁹⁰de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 160.

⁹¹de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 161.

gathered from wild rubber trees by contract workers. Although the exploitation of tappers by traders and barons was significant and the labor productivity of wild tapping was low, there were several factors that impeded the rationalization and industrialization of rubber production in Brazil. Scholars note two explanations for the relative failure of proletarianization among Brazilian rubber tappers. The first emphasizes the dispersed nature of the workforce alongside tapper resistance: the desire among tappers to maintain the relative autonomy of piece work and their aversion to wage-labor. Along with the traders, tappers actively resisted the rationalization of their production.⁹² The second explanation was the notorious South American leaf blight which had limited spread in the wild, but thrived on plantations with rows of close planted trees.⁹³ This disease, native to South America, was not present in Asia, and so had no deleterious effect on the development of plantations in Colonial Malaya.

As wild harvested rubber declined on the world stage, hampered by the South American blight in the Amazon, King Leopold's personal finances in the Congo, and the low productivity of wild rubber collection, it was replaced by plantation rubber in southeast Asia. Rubber plants were first brought to Malaya in the 1870s, but were not cultivated on a large scale until 1897-8 when plantation owners were looking for a replacement for failing coffee crops.⁹⁴ With the Malay Reservations Enactment in 1913, foreigners were allowed to purchase land in certain areas and large rubber plantations developed, requiring a large immigrant workforce. To supply this demand, the government of the Federated Malay States proposed a policy centered around

⁹²Barbara Weinstein, "Capital Penetration and Problems of Labor Control in the Amazon Rubber Trade," *Radical History Review* 27 (1983): 121-40.

⁹³Warren Dean, *Brazil and the Struggle for Rubber: A Study in Environmental History* (Cambridge University Press, 1987).

⁹⁴J. H. Drabble, "The Plantation Rubber Industry in Malaya up to 1922," *Journal of the Malaysian Branch of the Royal Asiatic Society* 40, no. 1 (1967): 54.

three points: the development of a free workforce; the recruitment of immigrant workers from India, and regulation of employment conditions (e.g. hospitals, water, housing etc.).⁹⁵

In practice, this policy led to significantly better working conditions than those in the Amazon and the Congo, but worse conditions than those of auto workers in Michigan. Between 1840 and 1940 in one of the largest migratory systems on earth some 28 million people crossed the Bay of Bengal, many of them low-caste Tamils looking to work on the rubber plantations in Malaya.⁹⁶ Their legal status fluctuated between indentured, debt-bonded, and free, but these labels often had little influence over their day-to-day working conditions. J. Norman Parker, in *Colonial Labor Policy*, notes that in Malaya, “to be a free laborer often had little real meaning since the working conditions of a free laborer and an indentured laborer differed little or not at all.”⁹⁷ Most of the workers, indentured or free, worked for a wage that was differentiated on type of job and the sex and race of the worker. Women generally received a fraction of their male co-workers wages, and Chinese laborers received more than Indian or Javanese workers. For all workers, the workday was limited, although planters struggled to extend it.

In Malaya, a steady supply of workers was ensured by the enticement of good pay, and “in the possibility of his being able, during his two or three years average stay in this country, to remit small amounts of money to relations.”⁹⁸ An interesting feature of the Malayan rubber industry is the recruiter/overseer called a *kangany*.⁹⁹ These individuals would travel back to India

⁹⁵Drabble, “Plantation Rubber Industry,” 59.

⁹⁶Sunil S. Amrith, *Crossing the Bay of Bengal: The Furies of Nature and the Fortunes of Migrants* (Harvard University Press, 2013), 2.

⁹⁷J. Norman Parmer, *Colonial Labor Policy and Administration: A History of Labor in the Rubber Plantation Industry in Malaya, Monographs of the Association for Asian Studies, IX* (J. J. Augustin Incorporated Publisher, 1960), 16.

⁹⁸Parmer, *Colonial Labor Policy*, 177.

⁹⁹Tirthankar Roy, “Sardars, Jobbers, Kanganies: The Labour Contractor and Indian Economic History,” *Modern Asian Studies* 42, no. 5 (2008): 971–98.

to recruit family members as rubber workers, and would later serve as their managers on the plantations, employing a combination of coercions based on debt and familial relations. As in the Congo and the Amazon, the labor force was not reproduced locally and generationally, but through tapping an external source of potential workers. In this case, the government of India worked with Malaya to facilitate the migration of workers from southern India to the Malayan rubber fields. Unlike in the wild cultivation fields across the world, workers in Malaya were paid in cash, and employers were directed to provide for workers' housing, sanitation and medical care in addition to wages.¹⁰⁰ In the 1920s, the level of acceptable wage and quality of social provisions was a fixed point of contention between the government of India, who facilitated the supply of immigrant workers, and the government of Malaya and the planters, who wanted to reduce costs.

The question of social reproduction is complicated by the introduction of migrant and indentured labor. As with seasonal/transhumant labor in the Amazon, the social reproductive dynamics of migrant workers in colonial Malaya were stretched across vast distances. The existing communities of workers in India and Malaya were disrupted by migration, but do not seem to have been destroyed as in the Congo and among indigenous Amazonians. Rather, the social reproduction of the workforce became bifurcated, operating locally within the amenities of the rubber plantations and regionally through cash transfers across borders. This introduces significant complications to the question of labor reproduction, but provides an exciting opportunity to examine the fundamental relationship between labor and capital as it is attenuated by geographical and political space.

¹⁰⁰Parmer, *Colonial Labor Policy*, 177-8.

A steady supply of laborers is a central problem for industry everywhere. The social-reproductive conditions of laborers varied widely along the global automotive commodity chain. Even at the point of the chain where rubber entered the production process, the living conditions of workers in different regions were structured differently by global and local pressures. The conditions of the reproduction of labor responded to factors such as the availability of cheap labor from an outside source, the resistance of workers to changes in their social and communal patterns, and the type of labor required by capital.

Conclusion

The concept of Fordism is useful in that it highlights several fundamental dilemmas facing capital at the turn of the twentieth century. The expansion of the production of commodified domestic goods was combined with an expansion in the market for their consumption. However, although the networks of consumption and production were global, the techniques used to effect their expansion were not uniform across the globe. As we see in the early automotive industry, capital employed several distinct methods of exploitation that corresponded to different cultural, political, and ecological conditions. Recent theoretical work on the household allows us to understand the role that diverse structures of reproduction play in this global network of production.

The primary question posed by this chapter is that of the articulation between capital's need for specific kinds of labor, and the specific ways in which it reproduces the corresponding different kinds of laborer. How did global configurations of social reproduction respond to the Fordist revolution in production? Following from the insights of Marxist Feminists, I posit that capital made use of different household patterns to structure the reproduction of labor, but this is

not a simple dynamic to investigate. It is clear that neither conditions of production nor reproduction were exported globally wholesale by Fordist producers, and yet the entire world of production and reproduction was certainly transformed.

In the Amazon and the Congo, the reproduction of the debt relation and of the tax relation is inextricable from the reproduction of labor. Although commodity consumption was introduced through the closed provision shops, household production was destabilized, rather than supported. The razing of homes by slaving expeditions in the Congo, and the “kidnapping of indigenous infants ... to be raised within the families of enforcers or patrons to work the rubber fields”¹⁰¹ in the Amazon are the logical parallels of the planned communities in Michigan. In Michigan a labor supply was ensured and renewed through the new family patterns of the middle class; in the Global South this reproduction of labor was ensured and renewed through the destruction of indigenous families.

The supply of labor on the Malayan plantations was also secured from external sources, but through economic enticement rather than brute force. Plantation workers were paid a wage, but as their housing was provided for by the planters, and the state enforced labor laws regarding the length of the workday and workweek. The tenure of migrant workers was short, and for many workers a portion of his/her wage was used to support families back home in India.

These patterns of reproduction are connected across the world-system by commodity chains. A finished car needed inputs of iron, wood, coal, and rubber, and each of these had to be gathered and prepared by laborers far from auto factories. The labor processes, and the conditions of its renewal, were conditioned by various local factors - the position in the global economy, the extant patterns of life and labor, and the family formation in which laborers lived.

¹⁰¹de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 151.

While factory workers in America could be said to be proletarians, slave laborers and informal workers in the rubber fields could not. And while production for the world market dominated the lives of workers and their families, it was not the only type of work they engaged in: they were supported by unique combinations of wage labor, unpaid reproductive work, and subsistence labor. If Fordism can be seen as a reconfiguration of the articulation between patterns of consumption and production, then the changes that we have traced across the Global South form a part of the regime of Fordism and warrant further study within this framework.

Chapter 2: Social Reproduction in Fordist Michigan

As we saw in the previous chapter, the Fordist revolution was a watershed moment in the history of capitalist production. It set, in the 1910s, a standard pattern of production that would be taken up across industries in the core regions of the capitalist world-economy. However, very few of the technical innovations attributed to Henry Ford actually originated in the Ford Motor Company. Sociologist Simon Clarke notes that “the decomposition of tasks, the specialization of tools, the assembly of tools into the machine, and even of machines into the machine system”¹⁰² were already common in US industry prior to Ford’s project – even the automated assembly line, Ford’s most notorious invention, was actually developed in 1911 by EMF Company, a rival automobile firm.¹⁰³ For Clarke, Ford’s real innovation was that he “applied these principles to a new branch of production ... with such single-minded ruthlessness that he transformed the conditions of production motor vehicles almost overnight.”¹⁰⁴

The ruthlessness with which Ford reconfigured the labor process was mirrored in his approach to controlling labor inputs. After the establishment of the moving assembly line and the consequent reconfiguration of the labor process, the primary remaining barrier to efficient production at Ford was employee turnover- 400% in 1913. In addition to training new workers hired to support growing production, Ford also was constantly training replacements for workers

¹⁰²Simon Clarke, “What in the F—’s Name Is Fordism,” in *Fordism and Flexibility: Divisions and Change*, ed. Nigel Gilbert et al. (Palgrave Macmillan UK, 1992), 16.

¹⁰³Wayne Lewchuck, “Fordism and the Moving Assembly Line: The British and American Experience, 1895-1930,” in *On The Line: Essays in the History of Auto Work*, ed. Nelson Lichtenstein and Stephen Meyer (University of Illinois Press, 1989), 20.

¹⁰⁴Clarke, “What in the F—’s Name Is Fordism,” 17.

who left. With training costs between \$35 and \$100 for each new worker, Ford was hemorrhaging money that could not be recouped by suppressing wages.¹⁰⁵ Ford's solution? Pay better. But not just better, a lot better. In January 1914, to the outrage of other regional employers, Ford doubled the salary of auto workers from \$2.50 to \$5.00 a day. The name of this program, the "Five Dollar Day," belies the scope of Ford's concern with labor input: workers were only eligible for higher wages if they met certain moral standards in their daily lives, to be confirmed by a team of sociologists hired by Ford to investigate his employees. Ford's management practices, as well as those of other automobile firms such as GM, intentionally blurred the line between work and private life: private life and the home came to be seen as an extension of the shop floor, crucial zones for the management of the relations of production and reproduction of capital.

There has recently been a wealth of scholarship on the social reproduction of the conditions of capital accumulation - both on a high theoretical level and in specific historical situations. The work of Marxist feminists as well as world-systems theorists have gone a long way in demonstrating the centrality of processes outside the wage relation to capital accumulation. The household especially is crucial as a site of both the reproduction and socialization of laborers, and as the most important site of commodity consumption. In this paper I will examine the specific conditions of social reproduction in Fordist Michigan. The configuration of working-class life underwent a profound change at the beginning of the twentieth century, and industrial Michigan was, if not necessarily the epicenter of this change, at least one of its clearest examples. As the tasks of production were divied up between

¹⁰⁵Thomas Klug, "Employers' Strategies in the Detroit Labor Market, 1900-1929," in *On The Line: Essays in the History of Auto Work*, ed. Nelson Lichtenstein and Stephen Meyer (University of Illinois Press, 1989), 54.

decomposed, rote tasks and the planning and supervision of engineers and managers, the working classes were similarly divided. High wages for some workers were established alongside a growing matrix of informal labor and unpaid domestic work. I am interested in the articulation of the relations of reproduction in this specific historical situation, and especially the stratified and segregated ways in which specific types of labor were produced and reproduced in industrial Michigan. In other words, I ask not merely how labor power was reproduced, but how the matrix of stratified concrete labors was reproduced.

In choosing to focus on industrial Michigan, I am influenced by both the Marxist geography of Michael Storper and Richard Walker, and by the insights of world-systems analysts, especially Giovanni Arrighi. In his analysis of the rise of the US as global capitalist hegemony, Arrighi credits two aspects of American capitalism: the “internalization within a single organizational domain of activities and transactions previously carried out by separate businesses”¹⁰⁶ – that is, vertical integration – and a new relationship with space. Crucial to the rise of US hegemony was a massive continent full of rich reserves of labor and raw materials. Accessing these materials required, alongside a genocidal campaign of dispossession of indigenous communities, a material transformation of the relationship between capital and space: “as soon as the Revolution had freed the settlers’ hands, they set out to conquer as much of the North American continent as was profitable and to reorganize its space in a thoroughly capitalistic manner.”¹⁰⁷

Industrial Michigan was central to both of these developments in US capital accumulation. It sat at the entrance to the American west, and after the opening of the Erie

¹⁰⁶Giovanni Arrighi, *The Long Twentieth Century: Money, Power, and the Origins of Our Times*, 2nd ed. (Verso, 2010), 247.

¹⁰⁷Arrighi, *The Long Twentieth Century*, 62.

Canal Detroit was transformed into a crucial shipping port. Prior to the rise of the automobile industry, Michigan was at the center of US production in ship engines, rail-cars, and lumber – all industries crucial for the development of land west of the Mississippi river. The rail industry in the late 19th century, and especially the auto industry in the early 20th were established on the principle of vertical integration as a way to minimize transaction costs and maximize production. Most notorious is Ford’s River Rouge plant which turned raw materials into cars within a single massive complex. For Storper and Walker, this was the “long wave of early Fordism” in which “the Midwest became the country’s industrial core led by the automobile ensemble centered on Detroit and the machine-tool complex focused on Chicago.”¹⁰⁸

My aim is to show that this historical shift in the relations of production was accompanied by very deliberate management of the conditions of the reproduction of the conditions of capital accumulation. This brings us to the central questions of this paper: what role does private life outside the firm play in capitalist production, and how has capital attempted to exercise control over it? To approach this question, we will look at the conditions of reproduction in industrial Michigan in the early decades of the Fordist revolution, paying special attention to the policies of both the Ford Motor Company in Detroit and its rival General Motors in nearby Flint. The Five Dollar Day, implemented by Henry Ford, and GM’s Modern Housing Corporation were both concerted efforts to reconfigure the relationship between public and private life, and to bring this relationship under the control of the capitalist firm.

We should note that the automotive industry in the US Midwest was not isolated from other industries or regions. Before the dramatic ascent of Ford and General Motors, many

¹⁰⁸Richard Walker and Michael Storper, *The Capitalist Imperative: Territory, Technology, and Industrial Growth* (Blackwell, 1989), 207.

regions of the US had thriving automotive industries. Many of these industries implemented social reproductive policies similar to those we find in Michigan - although usually on a smaller scale. Likewise, industrial firms were not the only agents in the reconfiguration of reproductive relations at the beginning of the twentieth century. A clear line can be drawn from the social-reproductive policies of Ford and GM and the welfare policies implemented by the federal government under Roosevelt. The reason we are interested specifically in Fordist Michigan is that the policies developed there prefigured later developments.

To begin this examination, we need a theoretical apparatus upon which to hang our facts and figures. We employ two distinct frameworks: the work of world-systems analysts and of Social Reproduction theorists, as elaborated in *Social Reproduction Theory*. Combining the insights of these two very different traditions allows us to identify the relationship between global dynamics of production and the local reproduction that takes place within working households. Bringing Social Reproduction Theory (SRT) to bear on our empirical study of labor reproduction in Fordist Michigan will have the secondary effect of allowing us to identify both the strengths and limitations of SRT. The next section will trace SRT from its roots in the work of Lise Vogel to its current iteration, and will consider several criticisms of that iteration, especially those posed by Kirstin Munro.

The Reproduction of Capitalist Society

The question of reproduction is central to the historical study of capitalism. In order to persist as a social formation, capitalism requires that certain conditions be met, and continually renewed. Some of these are basic to every society - as humans, we require the natural materials that form the stuff of subsistence, and we require human labor to work them into a usable state. It

is through this metabolism with nature that we can survive, but this relationship can be mediated by quite distinct social formations. This variety of social formations have been analyzed under the category 'modes of production', defined by Louis Althusser as "a way or manner (a mode) of producing. Of producing what? The material goods indispensable to the material existence of the men, women, and children living in a given social formation."¹⁰⁹ Althusser breaks down this category further into two constituent elements, the forces of production and the relations of production. The forces of production comprise: the *objects of labor*, such as raw materials or sources of energy; the *instruments of production*, such as tools or machines; and the *agents of production*, or the laborers who set upon the objects of labor with their instruments.¹¹⁰ Access to each of these three factors in the forces of production must be continually renewed; we must cultivate raw materials or find new sources where they can be extracted, we must maintain our machines and tools, and we must ensure a steady flow of laborers to work them. The structure of these factors - their internal relationships and the conditions of their reproduction - is effected by the relations of production.

Under the capitalist mode of production, there is a strict separation between the workers (agents of production) and the instruments and objects they employ in the production process. In other words, the laborers do not own the objects with which they labor - the tools and raw materials are owned by another party who stands outside the forces of production: the capitalist. This means, firstly, that production is controlled not by those who will make use of the commodities produced, but by those who are concerned only with profit. Capitalist production is commodity production, production for the market. It subordinates the creation of useful things to

¹⁰⁹Louis Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, trans. G. M. Goshgarian (Verso, 2014), 22.

¹¹⁰Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 25.

the creation of profitable things; it is a "mode of production whose overriding objective is to produce, not objects of social utility, but surplus-value and capital itself."¹¹¹ Secondly, under the capitalist mode of production, workers cannot directly labor to meet their own needs, but must labor for a capitalist in the production of commodities. In exchange for this labor, the capitalist who then pays them enough to survive, but less than the value of what they produce. This relation, the exchange relation, is the center point or determining element of what Alfred Sohn Rethel calls the 'social synthesis': "the network of relations by which society forms a coherent whole."¹¹²

What this means is that the reproduction of capitalist society is effected through exchange. On the one hand, this means that laborers are dependent on the act of exchanging for their own survival. They cannot grow food because they have no land, and they cannot eat the commodities they produce because they belong to the capitalist. They must resort to exchanging their labor for wages, and their wages for the objects of the labor of others. On the other hand, the other elements of capitalist society are also reproduced through exchange: the capitalist must purchase the raw materials and tools used in the production process, and the profit that is his *raison d'être* requires that the commodities he produces be purchased by some end consumer.

This requires some clarification, as these conditions are not universal within the capitalist world-system. On the one hand, Marx's 'so-called primitive accumulation,' the violent plunder of the Global South and the dispossession of producers from their land, seems to be not merely the starting point of capital, but an ongoing structural feature.¹¹³ The imposition of the exchange

¹¹¹Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 33.

¹¹²Alfred Sohn-Rethel, *Intellectual and Manual Labour: A Critique of Epistemology*, trans. Martin Sohn-Rethel, Historical Materialism Book Series (1978; Brill Academic Publishers, 2020), 4.

¹¹³William Clare Roberts, "What Was Primitive Accumulation? Reconstructing the Origin of a Critical Concept," *European Journal of Political Theory* 19, no. 4 (2020): 532–52.

relation as social synthesis is based on a regime of continuous violence.¹¹⁴ Furthermore, even within the core countries, forms of non-waged production persist. The enslaved laborers in American prisons are not wage-workers, neither are small farmers and agricultural producers, nor petty commodity producers and gig workers that work inside their homes. In what sense do these non-waged social configurations of production belong to the capitalist mode of production? This question has bugged scholars of capitalism who have come up with endless contradictory explanations. Althusser refers to them as "archaic forms" that have been dominated by capital "by virtue of the capitalist market's nearly total domination of the 'small independent producers'".¹¹⁵ In other words, although the empirical characteristics of non-waged labor seem to be holdovers from a "pre-capitalist" era, they have nonetheless been reoriented towards production for the market.

Jairus Banaji takes a similar approach, emphasizing the difference between modes of *production*, and modes of *exploitation*. In a discussion of what he calls vulgar Marxism, focusing on Maurice Dobb, Banaji critiques the view that "modes of production were deducible, by a relation of 'virtual identity', from the given forms of exploitation of labour."¹¹⁶ His critique of Dobb and the vulgar Marxists who posit a relation of identity between wage-labor and capitalism itself borrows from Lenin's analysis of the development of capitalism in Russia:

Lenin argued, in other words, that as simple-commodity producers are subordinated to the power of capital, in town or village, and specifically bourgeois relations of production develop, far from transforming bondage and serf-forms of exploitation into specifically capitalist forms of exploitation, i.e., those forms which correspond to the 'classical, adequate mode of production of capital', these new relations of production, founded on capital, intensify the existing backward forms of exploitation:

¹¹⁴Werner Bonefeld, "The Permanence of Primitive Accumulation: Commodity Fetishism and Social Constitution," *The Commoner* 2 (2001): 1–15.

¹¹⁵Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 29.

¹¹⁶Jairus Banaji, *Theory as History: Essays on Modes of Production and Exploitation* (Brill Academic Publishers, 2010), 53.

these forms remain 'feudal' or 'semi-feudal' in character, while the relations of production acquire a bourgeois character.¹¹⁷

The argument that the development of capitalism intensifies the so-called 'pre-capitalist' forms of exploitation is itself not new to Lenin, but is also found in Marx. For Marx wage-labor was not an isolated phenomenon, divorced from any other social configuration. Rather, "the veiled slavery of the wage-labourers in Europe needed the unqualified slavery of the New World as its pedestal."¹¹⁸ Crucially, the relationship between wage-labor in the core and 'unqualified slavery' elsewhere is not unidirectional, with unfree labor providing cheap inputs for production based on the wage. The development of capitalist relations transforms other types of exploitation: "as soon as peoples whose production still moves within the lower forms of slave-labour, the corvee, etc. are drawn into a world market dominated by the capitalist mode of production, whereby the sale of their products for export develops into their principal interest, the civilized horrors of over-work are grafted onto the barbaric horrors of slavery, serfdom etc."¹¹⁹ What emerges are new and more extreme configurations of slavery, serfdom, etc that did not exist before.

My issue with the above analyses is the characterization of non-waged labor formations as "archaic," "pre-capitalist," "backwards," or "lower forms." While it is true that the history of capitalism has led to an unprecedented expansion of wage labor, it also has created novel modes of exploitation based on extra-economic forms of coercion. To consign these to some sort of capitalist pre-history is to deny their specificity to the capitalist world, as well as to overstate the freedom of wage workers. Philip Corrigan writes that "[t]hese forms of coercion should not be conceptualized as residuals from former modes of production and essentially irrational; but as

¹¹⁷Banaji, *Theory as History*, 56.

¹¹⁸Karl Marx, *Capital: A Critique of Political Economy*, vol. 1, trans. Ben Fowkes (Pelican Books, 1976), 925.

¹¹⁹Marx, *Capital*, vol. 1, 345.

indications of capitalism's expansion."¹²⁰ In this context the thinkers we have cited are useful. The question is not why non waged forms of coercion have persisted, or if they will be swept aside by the progress of history, but how they are transformed and subordinated to the logic of capital: the nearly total domination of the capitalist market.

How is this discussion of modes of production relevant to the reproduction of capitalism? An analysis of social reproduction must look at the conditions for the creation and perpetuation of wage-labor, but it cannot be limited to this. It must include an analysis of the forms of unwaged labor that are internal to the capitalist world system, how these support and are supported by the system of waged labor. Furthermore, it must look at the conditions of accumulation and the domination of the market, how the conditions of this domination are effected and sustained. Keeping these complications in mind, this section will look at three aspects of social reproduction: the daily and generational reproduction of the abstract category of labor-power divorced from the means of subsistence; the reproduction of specific types of concrete labor; and the broader reproduction of market domination.

The Reproduction of Labor Power

The primary site of the reproduction of labor power is the household/family unit. The family, in its modern form, is not a transhistorical human organization, but a constituent institution of capitalist production. Immanuel Wallerstein, in his critique of traditional Marxist understandings of proletarianization, implicitly questions the highly idealized view of the family that would later be endorsed by some social reproduction theorists. Noting that capitalism

¹²⁰Philip Corrigan, "Feudal Relics or Capitalist Monuments? Notes on the Sociology of Unfree Labour," *Sociology* 11, no. 3 (1977): 441.

requires and expands the condition of the alienated wage-labor through the historical process of proletarianization, Wallerstein insists that throughout the globe the object of this process has not been the individual wage worker, but the household or family unit. Throughout history, “individuals have tended to live their lives within the framework of relatively stable structures which share a common fund of current income and accumulated capital.”¹²¹ These structures are generally (but not always) kin-based, and their boundaries are always shifting, in part due to the demands of the market: “It is in the context of such a household structure that a social distinction between productive and unproductive work began to be imposed on the working classes.”¹²² In this formulation, the modern family is not a “bulwark against capitalism” but is an artifact of capital, formed by the pressures of accumulation.

For Wallerstein, the capitalist family/household primarily serves as an income aggregator. As an income pooling unit, the household structures various forms of contribution to the means of subsistence for its members: for Wallerstein income refers to waged income, but also cash transfers, petty commodity production, rent, and subsistence activity. These five categories are combined in various proportions throughout the world, but generally wage-income is counterposed to the other four: Wallerstein considers households that depend primarily on wage-work to be proletarian households and considers those which depend primarily on other forms of income to be “semiproletarian households” (by far the larger group). Semiproletarian households are attractive to capitalists as “whenever wage-labor constitutes a substantial component of household income, there is necessarily a floor for how much the wage-earner can be paid.”¹²³

¹²¹Immanuel Wallerstein, *Historical Capitalism* (Verso, 1983), 23.

¹²²Wallerstein, *Historical Capitalism*, 24.

¹²³Wallerstein, *Historical Capitalism*, 35.

The semiproletarian household organization is promoted by capital in most of the world, as the matrix of non-waged income acts as a subsidy to the reproduction of labor: the developed reproductive sphere depresses wages. However, there are two forces resisting semiproletarianization. “One is the pressure of the wagedworkers themselves who seek to be ‘proletarianized,’ because that in effect means being better paid.”¹²⁴ Working families throughout the history of capitalism have preferred family structures that require less unremunerated reproductive labor and have struggled for this in unions and social movements. The second pressure comes from employers themselves, who have a “collective [long-term] need to have a large enough effective demand in the world-economy to sustain the market for their products.”¹²⁵ In Wallerstein’s conception, the family is a site of many conflicting tensions, but also as a structure that has been organized to aid capitalist accumulation.

Marxist feminists have long considered the ways in which the category of waged labor is reproduced and maintained by unpaid domestic labor within the household, often performed by women. In fact, it is difficult to analyze the category of labor power in modern society without also discussing the unique exploitation of women in its reproduction. Lise Vogel ascribes the phenomenon of the oppression of women in capitalist society to the role they play in the generational reproduction of laborers. In her 1983 book *Marxism and the Oppression of Women*, Vogel argues “for the power of Marxism to analyse the issues that face women today in their struggle for liberation” but she “rejects, however, the assumption made by many socialists that the classical-Marxist tradition bequeaths a more or less complete analysis of the problem of women’s oppression.”¹²⁶ Rather, she contends that the writings of Marx and Engels, as well as

¹²⁴Wallerstein, *Historical Capitalism*, 35.

¹²⁵Wallerstein, *Historical Capitalism*, 35.

¹²⁶Lise Vogel, *Marxism and the Oppression of Women: Toward a Unitary Theory*, Revised Edition, Historical Materialism Book Series 45 (BRILL, 2013), 2.

certain high-points in the twentieth century Marxist tradition, though they “never adequately addressed the question of women,” contain the kernel of a theory that can be developed into a “theoretical framework in which to situate the problems of women’s oppression and women’s liberation.”¹²⁷

Vogel identifies two dominant perspectives on the problem of women’s oppression, one which divorces women's oppression from the context of capitalism (the dual-systems perspective), and one which identifies women's oppression with their structural position in capitalist society (the social reproduction perspective). These approaches are basically incompatible. In the first approach, “women’s oppression derives from their situation within an autonomous system of sex-divisions of labor and male supremacy.”¹²⁸ In other words, the oppression that women in modern society face as women differs from the oppression that they face on the basis of class. There are obviously overlaps and intersections between the two, but logically, “the dual-systems perspective implies that women’s oppression follows a course that is essentially independent from that of class oppression.”¹²⁹ This has been the perspective of much radical feminism in the west, which has been at best agnostic towards anti-capitalism, and often hostile to it.

Vogel is more sympathetic to the social reproduction approach, although she considers it undertheorized. This approach grounds the oppression of women “in their differential position with respect to generational replacement-processes.”¹³⁰ This situates the analysis of the position of women in society within the “central dynamic of social development in societies characterized

¹²⁷Vogel, *Marxism and the Oppression of Women*, 2.

¹²⁸Vogel, *Marxism and the Oppression of Women*, 134.

¹²⁹Vogel, *Marxism and the Oppression of Women*, 134.

¹³⁰Vogel, *Marxism and the Oppression of Women*, 135.

by exploitation.”¹³¹ As capitalism is a total social system, transforming every aspect of modern life, a historically grounded critique of the oppression of women must take as a central concern the drive for endless accumulation. As the exploitation of labor-power drives accumulation under capitalism, the social-reproduction approach to the so-called women-question examines the unique role of women in the production and maintenance of labor-power.

However, Vogel points to a much more fruitful analysis of social reproduction. Her book deals extensively with the history of Marxist approaches to the liberation of women, but in the last chapters, she interrogates the question itself: “the first requirement for further motion is to abandon the idea that the so-called women-question represents an adequate category of analysis.”¹³² She elaborates the useful Marxist framework not as a theory of the position of women in society, but of the “reproduction of labour-power in the context of overall social reproduction.”¹³³ Within this dynamic, women occupy contradictory positions for capital – on the one hand, the potential capacity to produce future workers is necessary for continued accumulation in the long term, but on the other hand bearing children reduces a person’s capacity to perform productive labor for capital: “childbearing potentially entails a costly decline in the mother’s capacity to work, while at the same time requiring that she be maintained during the period of diminished contribution.”¹³⁴

Although her conception of ‘woman’ as simply those who have the capacity to bear children is perhaps dubious,¹³⁵ Vogel is making a broader point about gender and capital. For

¹³¹ Vogel, *Marxism and the Oppression of Women*, 135.

¹³² Vogel, *Marxism and the Oppression of Women*, 142.

¹³³ Vogel, *Marxism and the Oppression of Women*, 142.

¹³⁴ Vogel, *Marxism and the Oppression of Women*, 151.

¹³⁵ For a theoretical critique of Vogel’s transhistorical association of the category of woman with the process of childbearing, see chapter 7, “Capitalism and Difference,” in Søren Mau, *Mute Compulsion: A Marxist Theory of the Economic Power of Capital* (Verso, 2023). For a historical explication of the contingent nature of this association, see Maria Mies, *Patriarchy and Accumulation on a World Scale: Women in the International*

Vogel, women are oppressed in capitalist society due to their specific role in generational reproduction, i.e. the specific way in which they are dominated in service of the needs of capital. This consideration brings to the fore the articulation between productive labor and reproductive labor beyond just the question of childbearing. Reproduction entails not only the creation of new and future workers, but also the day-to-day maintenance of present workers and their dependents.

Vogel identifies this day-to-day reproductive labor as domestic labor. Though domestic labor is most often associated with feminized labor in private households, it “also takes place in labour-camps, barracks, orphanages, hospitals, prisons, and other such institutions.”¹³⁶ In other words, the oppression of women through the imposition of unwaged domestic labor is a specific manifestation of the larger phenomenon of reproductive labor within capitalism. Key to Vogel’s insight is that different forms of reproductive labor are utilized to reproduce different forms of labor. It is this larger phenomenon of social reproduction that has been elaborated in a recent trend in Marxist Feminism called Social Reproduction Theory.

Building on the work of Lise Vogel, Social Reproduction Theorists have staked out a formidable project: to reclaim fully half of the capitalist system as an object of study for Marxists. This project is elaborated in the recent celebrated volume *Social Reproduction Theory: Remapping Class, Recentering Oppression*, which defines Social Reproduction Theory (SRT) as “an approach that is not content to accept what seems like a visible, finished entity – in this case, our worker at the gates of her workplace – but interrogates the complex network of social processes and human relations that produces the conditions of existence for that entity.”¹³⁷ For

Division of Labour (Zed Books, 1986), especially chapter 3, “Colonization and Housewifization.”

¹³⁶Vogel, *Marxism and the Oppression of Women*, 159.

¹³⁷Tithi Bhattacharya, ed., *Social Reproduction Theory: Remapping Class, Recentering Oppression* (Pluto Press, 2017), 2.

the scholars working within SRT, this complex network of the relations of reproduction is crucial for the functioning of capital, and they are fundamentally relations of human labor.

This formulation is in direct contradiction to the self-conception of capital, which “acknowledges productive labor for the market as the sole form of legitimate ‘work,’ while the tremendous amount of familial as well as communitarian work that goes on to sustain and reproduce the worker, or more specifically her labor power, is naturalized into nonexistence.”¹³⁸ Indeed, Marx notes that value in capitalist societies is based not merely on labor-time, but on ‘socially necessary labor time’ an abstraction that elevates the importance of commodity producing labor and devalues all other types. The determination of what instances of labor are considered socially necessary is of course a contingent social process that responds to historical conditions. We will discuss this later in the paper, but for now it is enough to emphasize SRT’s insight that to the extent that reproductive labor happens outside the realm of commodification, it is “analytically hidden by classical economists and politically denied by policy makers.”¹³⁹

Social Reproduction Theory, then, is a theory of this invisible reproductive labor, and an attempt to understand it as constitutive of capitalism. As its state of ‘invisibility’ would indicate, this is far from a simple task. Bhattacharya notes the distinction between public/private to delineate reproduction from production. The life of the worker is divided into two sections: the working day during which the worker is engaged in the production of value, and the space of reproduction, in which the laborers reproduce themselves, or at least their labor-power. The distinction is not so cut and dry, however. These spheres, though “they may be separate in a strictly spatial sense are actually united in both the theoretical and operational sense. They are

¹³⁸Bhattacharya, *Social Reproduction Theory*, 2.

¹³⁹Bhattacharya, *Social Reproduction Theory*, 2.

particular historical forms of appearance in which capitalism as a process posits itself.”¹⁴⁰ In fact, production and reproduction are quite messily intertwined.

Quoting an earlier article by Johanna Brenner and Barbara Laslett, Bhattacharya defines social reproduction as inclusive of “how food, clothing, and shelter are made available for immediate consumption, how the maintenance and socialization of children is accomplished, how care of the elderly and infirm is provided, and how sexuality is socially constructed.”¹⁴¹ By this definition, reproductive labor includes the unpaid activity of mothers as well as waged nannies and daycare workers. It includes 'unproductive' teachers in public education ('unproductive' in the technical sense that the object of their labor is not a commodity) as well as 'productive' teachers in private schools (whose teaching *is* sold as a commodity). The network of reproductive relations is complex, pulsing just under the surface – unified by little other than its relation to labor-power and its devaluation by capital. It cuts across traditional Marxist understandings of labor: it includes both waged and unwaged work, it is sometimes commodified and sometimes not, it is performed by both men and women on both formal and informal bases, and it permeates all institutions of society from the home to the state itself.

The work of SRT scholars is at its strongest when it takes an expanded view not just of the types of labor required for capitalist accumulation, but of reproduction itself. This expanded view takes seriously Vogel’s consideration of the reproduction of labor-power that takes place in “labour-camps, barracks, orphanages, hospitals, prisons, and other such institutions.”¹⁴²

Reproduction is far more than mere biological reproduction, though that is a crucial aspect.

Labor-power is reproduced at several different levels of social organization – in addition to the

¹⁴⁰Bhattacharya, *Social Reproduction Theory*, 9.

¹⁴¹Bhattacharya, *Social Reproduction Theory*, 6-7.

¹⁴²Vogel, *Marxism and the Oppression of Women*, 159.

family, Bhattacharya notes that “[s]lavery and immigration are two of the most common ways capital has replaced labor.”¹⁴³ Societies can and have shifted the requirement to reproduce labor power from local families to the marginalized zones of the world, increasing the influx of workers to a region through invitation or theft. Additionally, the day-to-day reproduction of labor power in prisons and work-camps is performed not by family members, but by laborers themselves on an unwaged basis or by employees dedicated to reproduction, such as cooks or cleaners.

Capital has not, indeed cannot, commodify all aspects of reproduction. It may commodify aspects of care and subsume portions of the sphere of reproduction into production, but it requires as a foundation a matrix of unpaid and devalued life. Just what types of labor are commodified is historically contingent. Nancy Fraser’s essay in *Social Reproduction Theory*, “Crisis of Care” articulates this complex relation: “Historically, this work of ‘social reproduction’ has been cast as women’s work, although men have always done some of it too. Comprising both affective and material labor and often performed without pay, it is indispensable to society.”¹⁴⁴ Although the existence of this unpaid realm is crucial to the existence of capitalist accumulation, its boundaries are constantly shifting. In fact, its consolidation and denigration as “women’s work” has been a relatively recent (in historical terms) configuration: “Splitting off reproductive labor from the larger universe of human activities, in which women’s work previously held a recognized place, [industrial capitalist societies] relegated it to a newly institutionalized ‘domestic sphere,’ where its social importance was obscured.”¹⁴⁵

¹⁴³Bhattacharya, *Social Reproduction Theory*, 7-8.

¹⁴⁴Nancy Fraser, “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism,” in *Social Reproduction Theory: Remapping Class, Recentering Oppression*, ed. Tithi Bhattacharya (Pluto Press, 2017), 21.

¹⁴⁵Fraser, “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism,” 24.

Fraser positions this domestic sphere not as a universal category, but a contingent one. It is identical neither to unpaid work, nor to reproductive work, but is the lever by which the relationship between unpaid work and reproduction is manipulated. Fraser treats the categories of SRT – domestic labor, reproduction, the family, etc. – not as fixed units of analysis, but as categories to be critiqued. How, specifically, have the configurations of production and reproduction been molded by the demands of value accumulation, and how have those configurations been constitutive of historical modes of capitalism?

Fraser tracks the shifting configuration of social reproduction through three eras of capitalism: the liberal capitalism of the nineteenth century, the state-managed capitalism of the early and mid 20th century, and the present moment of neoliberal financialization. In each of these periods, the relationship between production and reproduction was manipulated in order to relieve the tension of a systemic crisis. Liberal capitalism preferred a free labor market where the reproduction of labor power in the capitalist core was offloaded onto families, as production and reproduction were forced into “separate spheres.” Drawing on Maria Mies notion of ‘housewifization,’ Fraser notes that this configuration of domestic labor in the core was dependent on the massive exploitation and devaluation of the labor of racialized women in the periphery.

After the ravages of the two great depressions in the 1870s-1890s and the 1930s, capitalist states reorganized the reproduction of labor-power on a model of state welfare. In this model, states assumed the burden of providing for reproduction, introducing in the US such measures as the minimum wage, social security, expanded public education and other programs of social protection and reproduction. State subsidization was limited, however. It was

systematically withheld from workers in the periphery who were again exploited in order to subsidize social reproduction in the core. For the current moment of neoliberalization Fraser notes that there is “a new, dualized organization of social reproduction, commodified for those who can pay for it and privatized for those who cannot, as some in the second category provide care work in return for (low) wages for those in the first.”¹⁴⁶ Racialized immigrant workers are often employed in these low-waged domestic jobs – once again supporting the reproduction of other classes, but in this case often in a commodified manner.

At its best, SRT is a powerful critique of the foundations of the capitalist system: the too-often unpaid and devalued work of reproducing labor power. The consideration of reproductive labor as constitutive of capitalist accumulation is an important corrective to masculinist and wage-based analyses of capital. Although it is true that waged labor is crucial to the accumulation of value, the unpaid work of women and racialized men (and especially the work of abject laborers in the periphery) form the conditions of possibility of waged work. The continual devaluation of certain types of labor, especially reproductive labor, is the historical foundation of capitalism.

Too often, however, instead of investigating the complex relationship between production and reproduction, social reproduction theorists have committed to an overly rigid distinction between the two spheres. Although noting that they are part of the same complex of social relations, Vogel insists that “[r]eproduction of labor-power is not, however, itself a form of production.”¹⁴⁷ Similarly, in Bhattacharya’s formulation: “SRT points out that labor power itself is the sole commodity ... that is produced outside of the circuit of commodity production.”¹⁴⁸ This

¹⁴⁶Fraser, “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism,” 32.

¹⁴⁷Vogel, *Marxism and the Oppression of Women*, 144.

¹⁴⁸Bhattacharya, *Social Reproduction Theory*, 7.

is only partially true. Bhattacharya is correct that the reproduction of labor power requires activity outside of commodity production, but capital has always been interested in the commodification of reproductive care work. Activities of production and reproduction are not distinct but overlapping, and separating them so sharply reduces our ability to theorize, the privatization of reproductive work. Capital, though required to offload some portion of the costs of reproduction onto unwaged workers or distant communities, has always retained a keen interest in the commodification of all life. When it has been able to afford producing the requirements of reproduction as commodities, it has done so.

This overly rigid circumscription of the reproductive sphere leads to some significant oversights in Social Reproduction Theory. Its biggest shortcoming is the occasional credulous endorsement of the family as an intrinsically radical space due to its presumed logical separation from production. But the family is not merely an institution that has been harmed by capitalism, but it is a fundamentally capitalist institution. When Mohandesi and Teitelman claim that “households have historically served as bulwarks against capitalism”¹⁴⁹ they accept at face value capitalism’s paradoxical glorification of the family as a sacred space, untouched by crass commercialism. In fact, the family was formed by crass commercialism, and it exists in (or rather, was disciplined into) its modern form to perpetuate the conditions of reproduction.

¹⁴⁹Salar Mohandesi and Emma Teitelman, “Without Reserves,” in *Social Reproduction Theory: Remapping Class, Recentering Oppression*, ed. Tithi Bhattacharya (Pluto Press, 2017), 40.

The Reproduction of Concrete Labor

In addition to its role in the generational and daily reproduction, the household has a secondary role as well. For Wallerstein, the household/family construct plays a vital role in capitalist society as the site of ideological reproduction. He argues that:

Households serve as the primary socializing agencies of the world-system. They seek to teach us, and particularly the young, knowledge of and respect for the social rules by which we are supposed to abide. They are of course seconded by state agencies such as schools and armies as well as by religious institutions and the media. But none of these comes close to the households in actual impact.¹⁵⁰

This socialization plays an important role in conditioning future (and present) workers to accept their role as either paid laborers or subsistence laborers, and “results in the internalization of the myths, the rhetoric, and the theorizing of the system.”¹⁵¹ Wallerstein notes that occasionally this socialization can have an anti-systemic orientation, as households often “socialize members into rebellion, withdrawal, and defiance,” however, “antisystemic socialization can be useful to the system by offering an outlet for restless spirits, provided that the overall system is in relative equilibrium.” Further, except in times of structural crisis, “negative socializations may have at most a limited impact on the functioning of the system.”¹⁵²

Socialization is not applied uniformly to workers in capitalist societies. The division of labor in society is reproduced by socialization. Even within the working class, there are division and demarcations between e.g. managers, supervisors, engineers and unskilled laborers. The boundary is not strictly impermeable - the occasional worker who rises the ranks functions as "a museum piece exhibited to encourage belief in the 'possibility' of the impossible and the idea that

¹⁵⁰Immanuel Wallerstein, *World-Systems Analysis: An Introduction* (Duke University Press, 2004), 37.

¹⁵¹Wallerstein, *World-Systems Analysis: An Introduction*, 37.

¹⁵²Wallerstein, *World-Systems Analysis: An Introduction*, 37.

there are no social classes...."¹⁵³ This is itself part of the ideological reproduction of the system, for "the 'technical' division of labour is quite simply a mask hiding the fact that some people are permanently 'penned' in the situation of the working class while others can have either high-level posts that are immediately attributed to them, or fairly or (very) broadly open ended 'careers'."¹⁵⁴ Socialization is crucial for maintaining the social division upon which the technical division of labor is based.

The household may be the main site of ideological reproduction, but it is not the only site. In an explicit critique of Social Reproduction Theory, Kirstin Munro looks beyond the household/family structure and emphasizes the potentially repressive nature of reproductive labor in general. Munro's critique of SRT is based in the claim that "Social Reproduction Theory overlooks the contradictory and antagonistic role of the state in the lives of people, as the reproduction of labor power in capitalism proceeds via antagonism and state repression."¹⁵⁵ Not only is the household itself a site of ideological reproduction, but the entire matrix of reproductive labor, whether performed for a wage or not, reproduces specific types of labor. Munro notes that the social role of reproductive and care workers such as nurses, teachers, and social workers "is the reproduction of not just any 'life' but that of a docile, exploitable worker."¹⁵⁶ When this labor is done by workers employed by the state, it represents not the state's interest in the well being of its subjects, but its interest in reproducing the specific types of labor historically necessary for the accumulation of capital.

¹⁵³ Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 37.

¹⁵⁴ Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 37.

¹⁵⁵ Kirstin Munro, "Unproductive Workers and State Repression," *Review of Radical Political Economics* 53, no. 4 (2021): 623.

¹⁵⁶ Munro, "Unproductive Workers and State Repression," 624.

Munro here focuses on non-household sources of reproductive labor: hospitals, schools, and social work organizations among others. These institutions are generally a part of the state, although private versions of them exist in growing numbers. Though considered ‘unproductive’ by capital for the reason that it does not produce commodities, the labor performed by workers in hospitals, schools and social work agencies serves a valuable role for the capitalist system: it disciplines the labor of others. Though the aid this work provides is necessary for the survival of many working class families, Munro highlights what she perceives as the complicity of these laborers in state repression. In particular, she rejects the characterization of the labor done by these workers as virtuous “life-making,” preferring a more complicated formulation that situates the type of life being made within “a larger process that perpetuates the capitalist organization of society.”¹⁵⁷ In other words, although the labor performed within hospitals, schools, and other reproductive institutions is essential for the renewal of life, the context in which it happens reinforces a repressive social structure. Indeed, “if exploitation and domination are inherent to the organization of production and reproduction in capitalism, then the reproduction of capitalist society proceeds by perpetuating these forms of organization.”¹⁵⁸ We see this play out explicitly in the continued perpetuation of structural racism in schools¹⁵⁹ and hospitals.¹⁶⁰

Munro’s critique of Social Reproduction Theory provides the impetus for the investigation central to this paper. While welcoming the achievement of SRT in its attempt to “overturn Marxist orthodoxy and shed light on the importance of unproductive labor carried out

¹⁵⁷Munro, “Unproductive Workers and State Repression,” 625.

¹⁵⁸Munro, “Unproductive Workers and State Repression,” 625.

¹⁵⁹Rita Kohli et al., “The ‘New Racism’ of K–12 Schools: Centering Critical Research on Racism,” *Review of Research in Education* 41, no. 1 (2017): 182–202, <https://doi.org/10.3102/0091732X16686949>.

¹⁶⁰Ellesse-Roselee L. Akre et al., “Measuring Local-Area Racial Segregation for Medicare Hospital Admissions,” *JAMA Network Open* 7, no. 4 (2024): e247473–e247473, <https://doi.org/10.1001/jamanetworkopen.2024.7473>.

by disproportionately women of all races and racialized men,”¹⁶¹ she rejects an uncritical view of the reproductive sphere and care-work as virtuous or revolutionary that is common (though not universal) in the writing of social reproduction theorists. In positioning the reproductive sphere as constitutive of capitalist accumulation and complicit in the repression of workers, she opens up the following question: How have the specific historical configurations of the reproduction of labor power – “the cultural norms that prescribe working class ‘needs,’ the specific proportions of inputs, the social assignment of tasks, and the arrangement of people in space”¹⁶² – been manipulated to fulfill the specific historical need of capital for certain types of workers.

We have pointed out that the claim by certain SRT scholars that labor power is produced “outside the circuit of commodity production” is only partially true. Indeed, Marx notes in volume 1 of *Capital* that “the ultimate or minimum limit of the value of labour-power is formed by the value of the commodities which have to be supplied every day to the bearer of labour-power, the man, so that he can renew his life-process.”¹⁶³ In other words, the value of labor power is, as for all other commodities, defined precisely by the extent to which it can be produced within the circuit of commodity production. This overlap between production and the reproduction of labor power is crucial because it represents an axis on which the value of labor power can be manipulated. If the value of the commodities required for renewing life-processes can be cheapened by the offloading of reproductive labor to unpaid or underpaid workers, then the value of labor drops as well.

The process devaluing labor by increasing the burden of uncommodified domestic work is not applied evenly to the working classes. Capital may value commodities based on an

¹⁶¹Munro, “Unproductive Workers and State Repression,” 623.

¹⁶²Munro, “Unproductive Workers and State Repression,” 624.

¹⁶³Marx, *Capital*, vol. 1, 276.

abstraction of undifferentiated labor power in various amounts, but it requires for the production process concrete acts of labor in various configurations and at various prices. Treating labor power as an undifferentiated mass may help us understand the logic of exploitation in a society based on generalized exchangeability - it is through the abstraction of 'labor-power' divorced from its immediate context that we can consider commensurate, for example, work done on a farm and that done in an office. However, in order to understand the historical configuration of production and reproduction in a given time and place, we must disaggregate this mass into its component parts. In other words, to build on Munro's critique, what is reproduced by domestic and unproductive laborers is not merely life, or labor power, but the specific kinds of concrete labor required by capital.

The Reproduction of the Market

Domestic and reproductive labor serves another crucial and much broader purpose for capital, in addition to its role in the reproduction of labor. The household is the primary institution responsible for renewing the life-force of commodity producers, but it is also the final destination of most of the commodities produced. Althusser writes that the family "is a *unit of consumption* in our societies. It is not the only kind of unit of consumption there is, but, of those in existence, it is a kind that still plays an extremely important part and it is not about to disappear."¹⁶⁴ Although many industries are involved in the production of machines or tools that are not directly consumed in private households, these can be seen as intermediate links in the commodity chains which terminate in the home: e.g. tractors and looms are built and sold as commodities, but are subsequently employed in the production of grain and cloth for clothes.

¹⁶⁴Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 77.

This grain and these clothes are then purchased and consumed in the household in what Munro calls the "household production process."¹⁶⁵ The household production process is not external to the circulation of capital, but is one of its main components.

The household production process is the final bit of processing that commodities undergo to make them fit for consumption: cooking food, cleaning clothes, employing household appliances such as vacuum cleaners in domestic chores, etc. This process combines two necessary elements: commodities purchased with a wage and unpaid domestic labor. For Munro, the household production process is linked to both the state and firms, as it "is forced to rely on capitalist firms and the state to ensure the survival of its members, both day-to-day and intergenerationally; it in turn contributes to the perpetuation of capitalist society."¹⁶⁶ In other words, we cannot isolate the household from broader capitalist society and identify it exclusively with the reproduction of labor. It must be theorized as part of a larger process: "[t]here is no social reproduction without 'societal reproduction,' as all production and reproduction in capitalist society are indelibly shaped by accumulation."¹⁶⁷ This means that households are complicit in the reproduction of the very conditions of their unfreedom. They are separated from the direct means of subsistence and forced to rely on commodity consumption in order to survive; in so doing, they valorize capital and provide the conditions of profitability that make endless accumulation possible.

This framework has important implications for the study of class struggle. If capital can make use of both devalued labor that is heavily dependent on unpaid reproductive work, as well as highly-paid labor that is supported by more extensive commodity consumption, then a 'win'

¹⁶⁵Kirstin Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," *Science & Society* 83, no. 4 (2019): 451–68.

¹⁶⁶Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," 454–455.

¹⁶⁷Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," 455.

for the working class is not necessarily a 'loss' for capital. Postone writes that "manifestations of class struggle between the representatives of capital and the workers over working-time issues or the relationship of wages and profits, for example, are structurally intrinsic to capitalism, hence an important constitutive element of the dynamic of that system."¹⁶⁸ The implication is that such struggles over the distribution of surplus are internal to capitalism: if the goal of class struggle is merely higher wages for the workers, or the ownership of a larger portion of surplus, then the success of the working classes means a decrease in the amount of capital available to be re-invested, but at the same time an expanded base of consumption through which the capitalist relations of production are reproduced.

The remaining sections of this paper will explore this tension in the context of industrial Michigan. As the Fordist revolution changed the way that capital is produced, it was accompanied by a revolution in the relations of reproduction. The relations of production were reproduced through transformed relationships between wage-labor, unpaid domestic labor, and commodity consumption. We will examine the extension by capital of new types of control over the private lives of workers in an attempt to tease out how specific configurations of household reproduction supported the new productive regime.

Households in the Michigan Automotive Industry

Prior to the Great Depression, automotive manufacturing was widespread in the US, especially in the Midwest and northeast although there were also firms on the west coast. However, the production innovations that originated in Ford plants and that were picked up and improved upon by firms such as General Motors or Ransom Olds were less widespread and after

¹⁶⁸Moishe Postone, *Time, Labor, and Social Domination: A Reinterpretation of Marx's Critical Theory* (Cambridge University Press, 1993), 36-37.

the Great Depression 50% of motor vehicle manufacturers had closed.¹⁶⁹ The Big Three automakers headquartered in Michigan - Ford Motor Company, General Motors, and Chrysler - emerged as the dominant forces in the industry. In this section we will explore the conditions of social reproduction in Detroit and Flint that contributed to their success during the first decades of the Fordist revolution. I highlight what I see as two important institutions through which industrial employers took control of the realm of private life – Ford Motor Company’s Five Dollar Day, and GM’s housing subsidiary the Modern Housing Corporation. The first is well studied, but we will be looking to the extent that the Five Dollar Day was generalized among Detroit workers, and the types of household formations that it influenced. The second is less represented in the literature, and so we will rely on a combination of archival documents and secondary literature. We will look at the time period stretching roughly from 1910, just three years before Henry Ford’s assembly line, to 1930 and the collapse of the global economy.

At the end of the 19th century and the beginning of the 20th, American industry underwent significant changes. During and just after the first world war, as industry was booming in the Midwest, waves of workers moved both north and west, whether Black workers looking for relief from the racial regime of the American south, or white workers from the northeast looking for new opportunities in the west. In the 1910s and 1920s, an estimated 1.5-2 million African Americans, predominantly men, migrated from the American South to cities such as Detroit, Chicago, New York, and Los Angeles in what is known as the Great Migration.¹⁷⁰ Although the Midwest, and Michigan in particular, had been a destination for immigrants looking for work since the late 18th century, the population virtually exploded in the twentieth.

¹⁶⁹Timothy Bresnahan and Daniel M. G. Raff, “Intra-Industry Heterogeneity and the Great Depression: The American Motor Vehicles Industry, 1929–1935,” *The Journal of Economic History* 51 (1991): 317–31.

¹⁷⁰Jr. Trotter Joe William, “The Great Migration,” *OAH Magazine of History* 17, no. 1 (2002): 31–33, <https://doi.org/10.1093/maghis/17.1.31>.

From 1910 to 1930, the population of Michigan almost doubled from 2.8 million to 4.8 million residents. Michigan's cities saw the greater portion of this boom as the Michigan economy shifted from extractive industries such as logging and copper mining to craft production and manufacturing. In these same two decades, Detroit more than tripled to over 1.5 million inhabitants, and Flint more than quadrupled to 150,000.¹⁷¹

Much of this growth was driven by two demographic sources: European immigrants (listed as “foreign-born whites” in the censuses) with high birth rates, and Black workers looking for better working conditions than were available to them in the American south. While the mass migration of willing (and as-yet un-unionized) workers was welcomed by employers, it created several real problems. There were simply more people moving to and living in the cities than could be comfortably housed with existing housing stock, and these new workers were unused to industrial work and the lifestyles expected of them.

The change in the nature of work and in the demographic composition of workers led to the development of new social institutions, as well as the rise of overcrowded slums in industrial cities. In the 1890s, Congress allocated funds for an investigation into the country's developing slums as a response to “civic leaders and voluntary associations [crying] out for action to improve the housing and surroundings of the ‘new Americans’ from Europe, and of the native poor.”¹⁷² This investigation was one of the first steps toward the development of city planning, and was part of a concerted effort “aimed at abolishing sweatshops, strengthening the labor movement, providing more manual and trade instruction, and improving education.”¹⁷³

¹⁷¹US Census Bureau, *Characteristics of the Population*, no. 2 (1940).

¹⁷²Mel Scott, *American City Planning Since 1890* (University of California Press, 1969), 7.

¹⁷³Scott, *American City Planning Since 1890*, 11.

The federal government would begin to play a larger part in providing for the welfare of its citizens during the Great Depression, with the bundle of social programs included in Roosevelt's new deal, but prior to that much of the work of dealing with the overcrowding and immiseration of urbanites was left to local and corporate actors. In Detroit in the 1920s, the "Detroit Bureau of Governmental Research, a private organization ... worked with a Mayor's Committee on Finances in estimating future increases in municipal revenues" in order to fund public services.¹⁷⁴ Even so, in 1927 the Detroit City Planning Commission and the Michigan Housing Association conducted "a series of surveys of Detroit and found 'what was to be expected' - that juvenile delinquency, deaths from tuberculosis and pneumonia, infant mortality, crime, and dependency were more prevalent in the blighted areas than they were in other parts of the city."¹⁷⁵ A major goal of the early automobile manufacturers was to insulate their workers from this blight, and so they developed private institutions of social control.

Immigrant workers coming from agricultural regions of the American South and Eastern Europe had to be actively conditioned to the industrial life of the Midwest. If Fordism was at its core the "transformation of production, according to strict criteria of technical rationality" then it requires for that transformation "new forms of social stratification, in strict accordance with the technical function, and corresponding new forms of morality and of personality, of socialization and education, to 'elaborate a new type of man suited to the new type of work and productive process'."¹⁷⁶ Gramsci refers to this social elaboration of a stratified society oriented towards the rationalization of work as 'Americanism' – imbricated with Fordism, but fundamentally distinct.

¹⁷⁴Scott, *American City Planning Since 1890*, 255.

¹⁷⁵Scott, *American City Planning Since 1890*, 256.

¹⁷⁶Clarke, "What in the F—'s Name Is Fordism." This phrase appears in the 1990 draft available at: <https://scispace.com/pdf/what-in-the-f-s-name-is-fordism-2bwbsf1e0m.pdf>

There are two aspects of Gramsci's concept of Americanism that are crucial to our present project – its historical specificity and its function as the mechanism of social differentiation. Although he welcomed the application of Fordist principles to European production, Gramsci was skeptical of the possibility of establishing Americanism in Europe as it requires a fundamental preliminary condition. "This condition could be called 'a rational demographic composition' and consists in the fact that there do not exist numerous classes with no essential function in the world of production, in other words classes which are purely parasitic."¹⁷⁷ European society, for Gramsci, is characterized precisely by the existence of 'parasitic' classes with no economic function beyond consumption. For Gramsci, it was precisely the lack of such a calcified social structure that allowed Fordism to succeed in America.

This brings us to the second point of interest in Gramsci's schema – Americanism is not a system of reducing stratification by the elimination of 'parasitic' classes, but is the creation of a new configuration of stratified society in which every social class is defined by its relation to production. For industrial Michigan, the massive waves of immigration in the early 20th century, although posing the problems of overcrowding and resource strain, provided the perfect opportunity for an experiment in Americanism. Families from Europe and individual workers from the American South, in looking for life and work, were funneled into different types of both. In our glance at the social patterns emerging in industrial Michigan, we will see that the different types of life created and reproduced correspond to the different types of work required by local industry.

¹⁷⁷ Antonio Gramsci, *Selections from the Prison Notebooks*, trans. Hoare Quintin and Geoffrey Nowell Smith (International Publishers Co, 1971), 281.

The Five Dollar Day – Patterns of Work

Enter the Five Dollar Day. As mentioned previously, the Five Dollar Day was a significant pay increase for skilled workers at Ford Motor Company designed to combat the region's high turnover rates. As most of the new laborers available to Ford came from agricultural backgrounds, training costs were so exorbitant that a 100% increase in wages was justified if it significantly reduced turnover. However, the Five Dollar Day was much more than just a raise. Half of the pay was categorized as a profit sharing scheme in which workers only qualified to participate if they met "specific standards of efficiency and home life."¹⁷⁸ As a part of this program, Ford employed a team of sociologists to ensure that employees spent their free time appropriately, and through his English school the Ford Motor Company "attempted to change an immigrant worker's life and culture to its preconceived ideal of an 'American standard of living,' which it felt was the basis for industrial efficiency."¹⁷⁹

Ultimately the Five Dollar Day was short-lived, interrupted by rising inflation and labor problems associated with the First World War. However, despite its early demise, it retains an outsized significance as an "early managerial strategy to match working-class culture to the requirements of the modern factory."¹⁸⁰ But the 'modern factory' is not the whole of production, and even within the factory, labor and pay were not evenly distributed by this program. Although there was little pay variation based on race within the Ford factories, Black workers were segregated by job type. They may have received similar pay to whites, but in order to do so they had to work in "hot, dangerous foundry jobs where quit rates were generally high."¹⁸¹ In other

¹⁷⁸Stephen Meyer III, *The Five Dollar Day: Labor Management and Social Control in the Ford Motor Company, 1908-1921* (State University of New York Press, 1981), 6.

¹⁷⁹Meyer III, *The Five Dollar Day*, 6.

¹⁸⁰Meyer III, *The Five Dollar Day*, 7.

¹⁸¹Christopher L. Foote et al., "Arbitraging a Discriminatory Labor Market: Black Workers at the Ford Motor Company, 1918-1947," *Journal of Labor Economics* 21, no. 3 (2003): 493.

words, Black workers were forced into the most precarious and dangerous jobs in order to preserve both the lives and careers of white workers. Even when Black workers received the same pay rate as their white peers, their labor was fundamentally ancillary.

The Five Dollar Day was never evenly applied to an entire population of workers. Kathleen Stanley and Joan Smith, in their study of the household in Detroit at the turn of the century, claim that “the crucial element of Fordism was the presumption of the so-called family wage – a male earned wage sufficient for the support of a dependent wife and children. However, in fact, even in Detroit, this presumed bastion of Fordism, the family wage was never really a reality for the majority of wage earners.”¹⁸² Rather, the majority of working families relied on supplemental income from wives and children, as well as significant subsistence labor. Moreover, after the economic collapse in 1929, working families depended increasingly on direct assistance from the state.

Stanley and Smith convincingly portray the position of households in Detroit as a textbook case of partial proletarianization. Although employers required wage laborers in increasing numbers for industrial production, they were loath to pay the full cost of that labor’s reproduction. And so the burden of reproducing labor at the level of the household was meted out to various actors. If we follow Marx’s analysis of the value of labor-power, then the Five Dollar Day, as it was granted only to portions of the working class, did not represent a general increase in the value of labor-power, but a new stratification of the value of labor-power. If the renewal of the life processes of some class of laborer can be supplied outside the commodity sphere by unpaid or devalued subsistence labor, then the value of labor-power for that class of laborer falls.

¹⁸²Kathleen Stanley and Joan Smith, “The Detroit Story: The Crucible of Fordism,” in *Creating and Transforming Households: The Constraints of the World-Economy*, ed. Joan Smith and Immanuel Wallerstein, Studies in Modern Capitalism (Cambridge University Press, 1992), 34.

This stratification is reflected in the structure of Detroit households. Stanley and Smith look at three types of families: US-born white families, Polish families (as representative of white immigrant workers), and Black families. In 1920, wives and mothers in white immigrant families were the least likely to work outside the home (about 1 in 20), and their children most likely. US-born white women worked at rates that were slightly higher, but “still quite low,” and their children were the most likely to stay in school. Stanley and Smith consider that the difference in family makeup between these two groups result from the higher wages of US-born white men, “72 percent of whom were either professional or skilled workers,” and because “Polish families brought with them skills associated their recent past as peasants” such as the practice of women keeping livestock and gardens.¹⁸³

While the differences in household structure between immigrant families, and the families of American-born whites were far from negligible, they paled in comparison to their shared differences with Black families. Unlike US-born whites with existing family structures in Michigan and white European immigrants, who usually migrated as family units, Black workers more often migrated as individuals without family or community contacts; this mass arrival of single Black workers was “a strong encouragement to the phenomenon of boarding, which helped both the new migrants and old residents to hold down housing costs.”¹⁸⁴ While boarding, or the inclusion of non-family members in a household, was extant among all ethnic groups, it was almost required in Black households as most Black laborers were unskilled and received wages far below the subsistence level. Boarding was, for Black households, a strategy of

¹⁸³Stanley and Smith, “The Detroit Story,” 38.

¹⁸⁴Stanley and Smith, “The Detroit Story,” 46.

survival. It was at once a result of the devaluation of Black labor in Detroit, and the conditions for the continuation of that devaluation.

Although boarding increased a household's cash flow via an extra wage-earner, it also required an increase in the demand of domestic subsistence labor, usually performed by women. Indeed, "in larger households, the addition of at least one woman boarder became necessary at certain points simply to share in the additional housework."¹⁸⁵ The increased domestic service load for Black women is illuminating when compared with white families within Detroit and Black families in other cities. Although married Black women worked outside the home at approximately half the rate within Detroit as Black women in other cities, they had outside work at five times the rate of immigrant whites: "In 1920, one in four married Black women worked for wages compared to only one in twenty married immigrant women."¹⁸⁶ Moreover, just as their burden of unpaid domestic labor was increased at home due to boarders, their waged labor outside the home was also domestic service. Black women were underrepresented in all occupations except service: whereas white women worked in not-insignificant numbers in manufacturing, trade, professional service and clerical work, 88% of Black women working outside the home in Detroit were employed as personal servants.¹⁸⁷ This is not to say that white women did not experience foreclosed employment opportunities relative to white men, but rather that the employment of Black women was overdetermined by both their race and their gender.

¹⁸⁵Stanley and Smith, "The Detroit Story," 47.

¹⁸⁶Stanley and Smith, "The Detroit Story," 46.

¹⁸⁷US Census Bureau, *Occupations, by States*, no. 4 (1930), 805.

Home and communal life

The working conditions of Detroit workers were inseparable from their living conditions. One of the most significant objects of Ford's sociological investigation, and prerequisite for increased pay, was workers' communal and private life. Sociologists working for the company would drive into working class neighborhoods and find neighbors, friends, and family members to vouch for the employee's character. Stephen Meyer notes that "the company held the worker responsible and accountable for the outside witnesses to his character. 'In cases where it is impossible to find someone who knows the habits and home life of the employee,' the home office instructed Ford branches, 'we withhold the increase [in profits] otherwise to be granted him under the plan'."¹⁸⁸ This would not necessarily be an issue for families with generations-long roots in the city, or for white European workers who migrated with their families, but for single Black migrant workers, or workers whose wives and children were forced to work outside the home, this amounted to almost an automatic disqualification.

Workers were also judged on their living conditions, with properly maintained homes factoring heavily into the judged fitness of a worker to receive higher wages. Meyer recounts a story of a German Catholic worker who was investigated by Ford's sociologists, and rejected from the profit share:

He lived in a two-story frame house that was in "poor condition." The neighborhood was also unacceptable: "Very bad. This is an alley with one and two story houses in poor condition on one side only. Foreign." The investigator detailed his impressions of the Ford worker's standard of living: "This man lives in a dirty unsanitary hut and has a room full of boarders, who sleep 3 and 4 to a room. Some of the boarders go through the room where the man and wife sleep to reach their own room. Wife looks haggard from overwork. She and the children are as dirty as their surroundings."

¹⁸⁸Meyer III, *The Five Dollar Day*, 130.

Living under such conditions, the investigator did not approve the worker for a share in the profits. Instead, he "advised" the worker not to have boarders, to move to a better neighborhood, and generally to improve his living conditions.¹⁸⁹

Even having boarders was considered a sign of unfitness, an inability to manage one's home. The worker in this anecdote eventually received his part of the profit share, but only when he had evicted his boarders, moved to the suburbs, and (on his own meager salary) built a free-standing single family home. Labor that is supported by certain household and reproductive configurations was explicitly devalued, which then reinforced the need for those configurations. This anecdote brings us to what I consider to be the most powerful factor in social stratification, and one of capital's greatest tools in disciplining the conditions of reproduction: housing, and its corollary, residential segregation.

Racial and residential segregation are extreme forms of social stratification. In industrial Michigan, residential segregation allowed for the housing shortage caused by increased immigration to be addressed on a differential basis – more of the limited resources available to cities could be directed towards housing more 'valuable' populations. Of course, racial segregation was not invented in 20th century Michigan – it had been a crucial element of colonial policy for hundreds of years. Carl H. Nightingale notes that the words "white town" and "black town," designating the sectors of a divided city, were first used by colonial city officials in Madras in South India.¹⁹⁰ This pattern was copied and reinforced by states and municipalities throughout the world for centuries, separating people of European ancestry from those they viewed as unequal. In the US, however, by the early 20th century, "black activists and their allies were able to convince the US Supreme Court to invalidate residential race segregation by

¹⁸⁹Meyer III, *The Five Dollar Day*, 133.

¹⁹⁰Carl H. Nightingale, *Segregation: A Global History of Divided Cities* (University of Chicago Press, 2012), 47.

government legislation.”¹⁹¹ The state did, of course, still pass racial legislation such as race-based immigration restrictions and lending policies, but it was no longer able to formally enforce the residential color line with direct legislation.

In order to work around this federal restriction, segregationists “reformulated the crusade for urban color lines in the United States. They culled a set of ideologies and city-splitting tools from the two other global-scale sources of institutional power used by urban segregation movements elsewhere – urban reform politics and the real estate industry.”¹⁹² This allowed for the creation of overcrowded slums to house poor and non-white residents, separated from the well-served affluent neighborhoods of skilled white workers and managers. In industrial Michigan, the institutions of segregation were developed by private industry, providing the state with a blueprint to continue the maintenance of a stratified society through indirectly selective welfare.

From 1910 to 1930, as Detroit more than tripled in size, the Black population grew from 5,741 to over 120,000 – a 20x increase. Many of these workers had lost jobs with the mechanization of southern agriculture and were moving to the Midwest, where manufacturing was booming. However, what awaited them was a potent combination of second-rate jobs and derelict neighborhoods. Mark Jay and Philip Conklin note that Black workers were forced into the “worst paid, most degrading, and most dangerous work Blacks were largely confined to jobs as production assistants, janitors and foundry workers.” This was reflected in housing, as “African Americans were also forced to live in the segregated Black Bottom neighborhood, which contained the city’s worst housing stock, often owned by predatory absentee landlords.”¹⁹³

¹⁹¹Nightingale, *Segregation*, 296.

¹⁹²Nightingale, *Segregation*, 296.

¹⁹³Mark Jay and Philip Conklin, *A People’s History of Detroit* (Duke University Press, 2020), 84.

It is well known that Black workers were forced to live in a state of segregation and abjection, and that they were often forced into bad jobs or paid less than their white peers. However, the connection between these phenomena is more complicated than it initially appears to be. Analysis of Detroit, both its history and its present, often centers a two-city hypothesis – one Detroit is a slum, and the other a great economic boomtown. However, Jay and Conklin argue that the “‘Two Detroits’ consensus in fact represents two components of a dialectical unity: redevelopment and austerity are not distinct processes but *two elements of the same process of uneven development*.”¹⁹⁴ They are writing of the recent gentrification, but the divided city has always been a complete, if complicated, whole.

Looking at the relationship between housing and demographic trends in employment, we begin to see how segregation served city employers, and particularly how residential segregation assisted in the process of Americanization. In Detroit in 1930, Black men were significantly underrepresented in trades and in professional service (architects, doctors, lawyers etc.) relative to their population, and significantly overrepresented in domestic and personal service. In manufacturing as a whole, Black workers were employed consistent with their relative population, but were largely absent from highly paid or skilled positions such as foremen, managers, or even machinists. Instead they were overrepresented in low-paid laborer positions: over a third of Black male industrial workers were employed as unskilled laborers for automobile factories, as opposed to less than 10% of US-born whites. When we speak of the socialization of new workers, it is clear that they were being socialized into different strata of the working class, separated largely by their race. This separation was reinforced and reproduced by housing and household patterns.

¹⁹⁴Jay and Conklin, *A People's History of Detroit*, 14.

The Modern Housing Corporation

In nearby Flint, Michigan, General Motors took a different approach to Americanism than Ford's Five Dollar Day, one that more explicitly emphasized housing as crucial to the elaboration of social stratification. This approach centered around the Modern Housing Corporation, a subsidiary of GM that purchased lots of land, developed them with houses and amenities and then sold them directly to employees. According to a retrospective report, written by a "Mr. Cowdrick" in 1930, the Modern Housing Corporation was created to conduct "all the more important housing activities" of GM.¹⁹⁵ These activities included the acquiring or building of more than 4,000 homes and 54 non-residential buildings, the vast majority in Flint, Michigan, but a small number also in Pontiac, Lansing, and Janesville, Wisconsin. The project was initiated in 1919 to combat limited housing due to "the expansion of automobile manufacture in the years immediately following the World War," and ran until it was dissolved in 1936. It seems to have always been viewed as a short term project to resolve a discrete problem, rather than a long term undertaking, as Mr. Cowdrick notes in 1930 that it "is expected that when all the properties have been liquidated, probably about 1945, all expenses will have been paid and the company will have recovered all the money it put into the project, but without interest."¹⁹⁶ That the corporation was liquidated 9 years before Mr. Cowdrick's prediction hints at who was expected to continue the project: the global economic crisis of the 1930s saw the state take over the management of the social status quo to an unprecedented degree.

Indeed, direct return on investment seems to have been less of a concern for the Modern Housing Corporation than exercising control over their employees' living conditions. The small

¹⁹⁵Cowdrick, *Operations of Modern Housing Corporation* (General Motors Library, 1930), 1.

¹⁹⁶Cowdrick, *Operations of Modern Housing Corporation*, 4.

profit made on each house in the project's mature years mostly funded the salaries of project administrators. After taxes and interest, if the company is to be believed (which is a big "if," to be sure), it "about '[broke] even' on its expenditures."¹⁹⁷ For the first several years of the project homes were sold at a loss, and for the length of the project, "at the time a house [was] sold, the Modern Housing Corporation ... secure[d] a mortgage from an outside lender,"¹⁹⁸ reducing the burden of capital outlay, but also significantly reducing the potential financing profits.

If the only problem was a housing shortage, then there were far more simple and far more profitable solutions than the Modern Housing Corporation. But when we understand the goal to be parallel to Ford's desire to "elaborate a new type of man," – or, drawing on Gramsci, to elaborate a new type of stratified society – then the efficacy of the Modern Housing Corporation is obvious. General Motors saw in the rapid growth of the auto industry and the rapid growth of Flint an opportunity to completely restructure the conditions of reproduction for life and capital. Flint's population was about 13,000 in 1900 and increased more than tenfold in thirty years. This forced the city to consider where and how to house the new workers, but more importantly, it had to decide what kind of city it would become. Tied up with this was the question of who would make the decisions about Flint's future, and who would provide the capital to fund the necessary development? The obvious answer seemed to be that Flint's biggest employer would fill both of these roles, and General Motors took to the task with enthusiasm. This was not just a chance to build houses for workers, but a chance to fundamentally shape every aspect of their lives.

General Motors did not create its housing division solely for the purpose of alleviating population issues, nor was the Modern Housing Corporation a venture into conventional real

¹⁹⁷Cowdrick, *Operations of Modern Housing Corporation*, 4.

¹⁹⁸Cowdrick, *Operations of Modern Housing Corporation*, 3.

estate investment. The project was explicitly one of Americanization, in which the reproductive apparatus was brought under the company's control. Mr. Cowdrick's report hints at the scope of the project. The homes sold to employees were not placed outside of cities in small company owned communities, but were inside city limits. This provided the communities with "the benefits of customary municipal services, including police and fire protection and sanitary regulations,"¹⁹⁹ at little to no cost to General Motors, but it also gave GM enormous leverage over the geography of the city itself.

Framed as General Motors' cooperation "with the public authorities for the benefit of the homeowners,"²⁰⁰ the Modern Housing Corporation's community administration plan was a collaboration with the municipality to allocate city resources to serve GM employees. The effect was to establish a standard pattern of reproduction embodied by a neighborhood of homeowners and then to transfer the responsibility for the reproductive labor of street and neighborhood maintenance to other entities. GM would make initial investments into public infrastructure that would then be taken over either by the city or by homeowners themselves. After an early contractor paid for the initial pavement of "about ten miles of streets ... a considerable amount of additional paving has been laid at the expense of the property owners." Additionally, the corporation planted trees along the streets, and noted that "home owners generally seem to be interested in improving their property and keeping it in good condition."²⁰¹

General Motors' administration of the Modern Housing Corporation showed a strong commitment to ensuring that employees had access to benefits far beyond mere housing stability. The corporation donated or purchased land for public schools, effectively choosing who would

¹⁹⁹Cowdrick, *Operations of Modern Housing Corporation*, 5.

²⁰⁰Cowdrick, *Operations of Modern Housing Corporation*, 5.

²⁰¹Cowdrick, *Operations of Modern Housing Corporation*, 5.

have ready access to education: “At Flint, there is one grade school and one junior high school inside the Modern Housing development Another grade school is contemplated, and will be located inside the Modern Housing development.”²⁰² To add to this, schoolteachers were one of the few non-employee groups who were allowed to purchase homes in the Modern Housing developments. By investing more in education for the employees who lived in the development, GM exercised control over just what type of workers their families children were being raised as. In addition to considerations for education, the corporation “laid out and improved a golf course and presented it to the city. It also erected a community building for donation to the city.”²⁰³ The golf course and community center were both public and although they were built by the corporation, they were maintained at the expense of the city.

From the first five pages of Mr. Cowdrick’s report, Flint seems like the apogee of the American dream: well-paid workers living with their families in owned homes on tree-lined streets; readily accessible public parks and community centers; schools and communities where teachers are given privileged status. One can imagine why Flint’s population quadrupled from 1910 to 1930. But as we know, this dream was built on uneven ground. The last paragraph of the report starts: “The regulations provide that houses may be sold only to members of the Caucasian race.”²⁰⁴ Just as in Detroit, the project of Americanism was not the creation of a more affluent class of workers, but the reconfiguration of a stratified working class.

Whites, especially well-employed whites, were encouraged to live in the well-served company communities, but GM did not create designated neighborhoods for Black and other non-white workers. They were segregated by default, by exclusion. This racial differential in

²⁰²Cowdrick, *Operations of Modern Housing Corporation*, 5.

²⁰³Cowdrick, *Operations of Modern Housing Corporation*, 5.

²⁰⁴Cowdrick, *Operations of Modern Housing Corporation*, 6.

attention (and funding) provided to their workers neighborhoods is a clear example of the reproduction not just of labor, but of differentiated laboring classes. Although the Modern Housing Corporation was explicitly developed to address the city of Flint's inability to house its growing population, it did nothing to help the fastest growing demographic: Black workers. While the white population almost quadrupled from 1910 to 1930, the Black population increased more than fourteen-fold. By default, these Black migrants looking for a place to live were consigned, along with poor white families, to "informal working-class settlements, most of which lacked urban amenities such as parks, sewers, and paved roads."²⁰⁵ Described by the city building inspector as "shanty towns," these unserved and informal suburbs were "more gritty and undeveloped than those in many other metropolitan areas."²⁰⁶ These neighborhoods were simultaneously the poorest, most racially diverse, and least served by city resources.

General Motors' project of Americanism was not a uni-directional imposition of norms: rather the makeup of the town and the makeup of the company were co-constituted. The Modern Housing project was a project of control over the conditions of capital's reproduction, both the reproduction of labor, and the realization of value in commodity consumption. GM sold houses to their best men, and then praised the men for their responsible homeownership, stating that "theoretically, foremen and superintendent do not favor customers of the Modern Housing Corporation in selecting men for retention, when reductions in force are necessary, but it is probable that the fact of home ownership exercises some influence."²⁰⁷ Even when a homeowner lost their job, they were given support by GM to keep their homes until they could resume making mortgage payments. The implication is clear: the valued worker is a homeowner, and a

²⁰⁵ Andrew R. Highsmith, *Demolition Means Progress: Flint, Michigan, and the Fate of the American Metropolis* (Chicago University Press, 2015), 28.

²⁰⁶ Highsmith, *Demolition Means Progress*, 28.

²⁰⁷ Cowdrick, *Operations of Modern Housing Corporation*, 6.

homeowner makes for a good worker. But this positive assertion of the value of a worker and their home life was predicated on the devaluation of a large mass of unskilled labor – and the consequent degradation of Black and poor neighborhoods. As GM funneled city funds and services to their Modern Housing developments, ultimately it was marginalized families that paid the price.

Conclusion

What emerges from this soup of demographic data is a complex hierarchy of labor that does not fit neatly into categories of waged/unwaged and productive/unproductive. A given type of work could fit into any combination of the above categories depending on who was performing it. Housework is unwaged and unproductive labor if done by a housewife to feed her family, or by a worker in their free time, but it becomes both waged and productive if performed on a waged-basis by a domestic servant hired on the open market. The physical act of the housework is identical in either case; the only change is the social context in which it is performed. The same could be said for cooking meals, or growing food, or teaching: it is not the act itself, but the social context of the act that determines its relationship to capital accumulation.²⁰⁸

In all of these scenarios, whether productive or not, the labor is reproductive. A worker needs clean clothes, a full belly, and skill in order to “arrive at the doors of her place of work every day so that she can produce the wealth of society.”²⁰⁹ Sometimes these necessities are

²⁰⁸ For the question of whether service can be considered a commodity, and therefore whether service work can be considered productive labor, see Michael Heinrich, *An Introduction to the Three Volumes of Karl Marx's Capital*, trans. Alexander Locascio (Monthly Review Press, 2012), 44. A service provided for pay is simply a commodity whose production and consumption are not separated either spatially or temporally: they occur at the same time.

²⁰⁹ Bhattacharya, *Social Reproduction Theory*, 1.

provided by the market, sometimes they are extracted from unpaid laborers, or from the devalued work of racialized communities. That the requirements of the reproduction of labor cannot be so easily disaggregated from the results of production does not mean that the sphere of “social reproduction” is incoherent. Social reproduction is a crucial category of analysis precisely because it is not simply the negative space of production: it is both a requirement of capitalist production, and a lever by which capital can reconfigure production inputs.

Historically the burden of unremunerated reproductive labor has fallen disproportionately on a gendered and racialized other. However this is not an absolute imposition, and the extent to which racialized and gendered proletarians have had to labor for a diminished wage or no wage at all is a historical question – one that we have tried to answer for early 20th century industrial Michigan. We will briefly recapitulate the results of this study below.

Fordist production, though often considered to be a process of deskilling labor, was in fact a complex reorganization of the relationship between labor and skill. A large portion of the laboring force was assigned decomposed tasks, most of whose required skill resides in complex machinery as dead labor. In a small number of positions such as engineering, managing, or the manufacture of engines, a large amount of skill was required. However most jobs were relatively low-skilled positions such as janitorial work, jobbing or assembly. This stratification did not exist only within individual firms – the increased stratification of the working class is observed across society. As the cities of industrial Michigan grew, the types of available jobs grew as well. A very small portion of workers were considered worthy of a family wage, but most workers survived by combining multiple incomes, or by supplementing substandard incomes with large amounts of subsistence work.

This should not be seen, however, as a simple case of cause-and-effect. Different family and household configurations did not develop in response to set wages. Rather, the larger employers in industrial Michigan systematically created a stratified society by exercising control over the private lives of workers. In the case of the Five Dollar Day, the Ford Motor Company required changes in the configuration of households before increased wages would be given. In order for a worker's labor to be deemed valuable, the worker's family had to disengage from patterns of informal subsistence labor. If they failed to do so, i.e. if the reproduction of their ability to labor continued to be subsidized by informal and non-waged income, then their labor was bought at a lower rate.

In Flint, General Motors took a more direct approach to structuring the private lives of their employees. Through the Modern Housing Corporation, they manipulated not only the family structures associated with higher-valued employees, but the very geography of the city. They used strategic investments to develop patterns of city life. By influencing the location of parks and schools, the planting of trees, and the paving of roads, they structured the reproductive environment of labor, answering Bhattacharya's questions "Does the education [the worker] received at school also not 'produce' her, in that it makes her employable? What about the public transportation system that helped bring her to work, or the public parks and libraries that provide recreation...?"²¹⁰ with an emphatic yes.

But not only the public schools, transit, parks and libraries – private institutions contribute to the reproduction of labor as well. If the domestic labor of a wife or child depressed the value of their husband's or father's labor power, that same work performed by any of the 88% of working Detroit Black women employed in domestic service had the opposite effect:

²¹⁰Bhattacharya, *Social Reproduction Theory*, 1-2.

now as a commodity contributing to the reproduction of the worker it contributes also to the value of his labor power. In Michigan, the value of domestic and ancillary workers' labor power was systematically depressed by the necessity of subsistence labor in underserved neighborhoods. At the same time, Black and female workers in reproductive jobs propped up the value of what industry considered to be the most important labor – that of white, male, and skilled workers.

The creation of a particular configuration of stratified society is a totally different thing than its maintenance. In this instance, capital and its representatives created certain conditions of social reproduction, but were quick to offload the responsibility at the first opportunity. General Motors transferred the responsibility for paving streets and planting trees to homeowners, and the running of schools and parks to the city of Flint. The management of wage-rates slipped out of the hands of employers with the development of a federal minimum wage as a part of FDR's New Deal, and the rise of a unionized auto industry with the Flint sit-down strike of 1937. It is important to note that both of these institutions, both early 20th century unions and New Deal programs, were more than happy to maintain segregation and a stratified workforce.

The point of this paper is not to directly link a certain type of reproductive scheme with Fordist production. As Gramsci noted, Americanism and its particular configuration of social life would be neither desirable nor possible outside of America. Rather I have attempted to demonstrate ways in which social-reproductive institutions are integrated into schemes of production. The specifics of this integration will change according to the needs of capital, but Social Reproduction Theory provides an excellent, if limited, starting point for their excavation.

Earlier I noted that a full elaboration of industrial Michigan's role in the United States' hegemonic ascent was outside the scope of this paper. The gap here is larger than I let on, primarily because 'industrial Michigan' is not a neatly bounded object – it is an incoherent object when divorced from the larger structure of the world-system and its globe-spanning commodity chains. Ford's River Rouge plant may have turned raw materials into cars in a single location, but the raw materials did not appear spontaneously at the factory gates. They had to be extracted from the earth by laborers outside of Detroit and shipped to the factory. A continuation of the line of inquiry expressed in this study will have to look along these commodity chains to truly account for the interlinked relations of production and reproduction supporting early Fordist production. This commodity chain is global, and includes the tens of thousands of workers in the rubber fields of South America, Africa, and Colonial Malaya. The study of productive and reproductive relations in the core is incomplete without an accounting of their co-constitution with similar relations in the global periphery. It is in this direction that the following chapters must turn.